

NEW CONSTRUCTIONS, OBSTRUCTIONS, AND MULTIPLIER STRUCTURE FOR ERDŐS PROBLEM 634

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ABSTRACT. Let $\mathcal{S}$ be the set of integers $N > 1$ for which some nondegenerate Euclidean triangle can be tiled by N congruent triangles. This paper continues the determination of the prime members of $\mathcal{S}$ and develops a framework for the full Erdős Problem 634. We give exact-coordinate certificates proving $88, 189 \in \mathcal{S}$, and an exact computer-assisted exhaustion proving $33 \notin \mathcal{S}$ whose complete branch tree is exported as a finite refutation certificate and validated by an independently written checker. The 189-tiling settles the first unknown multiplier on the Group 1 ray $21n^2$ and, with known constructions, shows that every multiplier $n \geq 2$ occurs on that ray.

For a primitive 120° tile with sides $c^2 = a^2 + ab + b^2$, we introduce a Laurent-polynomial boundary invariant whose quotient is $\mathbb{Z}/(3ab)$. It subsumes the ideal-membership obstructions from the known signed boundary characters and gives the exact arithmetic necessity on all six Group 2 target shapes. We also prove that every Group 2 multiplier set is a numerical semigroup with an explicit finite generator window. A trapezoid construction improves known sufficient thresholds; exact positive certificates and a one-implementation exhaustive search determine the entire one-leg family for the tile $(3, 5, 7)$. Supporting transfer and ordered-segment results isolate, but do not solve, the remaining geometric seam-propagation problem. Detailed computational qualifications, a result ledger, and a provisional programme below 200 are separated into the appendices.

1. INTRODUCTION

An integer $N > 1$ belongs to $\mathcal{S}$ if at least one nondegenerate triangle admits a dissection into N pairwise interior-disjoint triangles congruent to one another; reflections are allowed and the dissection need not be edge-to-edge. Erdős Problem 634 asks for a characterization of $\mathcal{S}$ [8, 14]. Classical constructions give

$$n^2, \quad 2n^2, \quad 3n^2, \quad 6n^2, \quad n^2 + m^2 \in \mathcal{S}$$

in their natural ranges [13, 14]. Recent classification theorems of Laczkovich, Beeson, and Zhang reduce all remaining possibilities to a small number of geometric branches [11, 12, 6, 7].

This paper is a sequel to [10]. Three contemporaneous manuscripts independently obtained the prime classification

$$p = 2, \quad p = 3, \quad \text{or} \quad p \equiv 1 \pmod{4}.$$

They use complementary direct and classification-based arguments [9, 10, 5, 4]; no claim of exclusive priority is intended. Beeson's four scalene 120° count formulae also confirm four rows of Table 1. These manuscripts settle only the prime subproblem; the full characterization of $\mathcal{S}$ remains open.

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The full problem is substantially harder than the prime case. Arithmetic classification places a candidate on a square ray $N = Cm^2$, but it usually does not decide which multipliers m are geometrically realizable. The three principal contributions of this paper are:

- (1) exact, independently checkable certificates for the new counts 88 and 189, together with an exclusion of 33 whose complete search tree is exported as a finite refutation certificate and validated by an independently written checker;
- (2) a single Laurent quotient that closes the $\chi(j, m) = (-1)^j z^m$ ideal-membership programme for 120° tiles; and
- (3) additive and finite-generation structure for the multiplier sets, turning every Group 2 ray into finitely many exact tiling questions.

Exact trapezoid, transfer, and ordered-segment results support this programme and identify the remaining geometric obstruction.

Evidence levels. A result labelled *proved* has a mathematical proof here or in a cited source. An *independently certified computer-assisted theorem* depends on an exact exhaustive search whose complete branch tree has been exported as a finite refutation certificate and accepted by a checker written from the certificate specification alone, sharing no code with the search (Appendix A.3). For such a result the search program is not part of the proof surface: the certificate, the checker, and the lemmas of Appendix A.2 are. A plain *computer-assisted theorem* depends on an exact exhaustive search for which no such certificate has been exported, so it rests on one implementation, and its completeness argument and reproducibility data in Appendix A are the proof surface. A *verified certificate* is a positive tiling rechecked from exact coordinates, independently of the search history. The label *conditional* is reserved for consequences of the unfinished forced-layer lemma in Appendix F. No conditional statement is used to prove an unconditional result.

2. THE CLASSIFIED BRANCHES AND THE PRIME CASE

We use the classification in the same form as [10], but we now state the imported statements on which every arithmetic branch of this paper rests. Throughout, T is the tiled triangle and R the tile, with angles α, β, γ opposite the sides a, b, c ; *commensurable angles* means that all three tile angles are rational multiples of π ; reflections are allowed and no tiling is assumed edge-to-edge. All statements below are quoted or transcribed from the cited versions; the version pins and the logical role of each are collected in the dependency ledger, Tables 7 and 8.

2.1. Imported classification statements. The classification splits by the shape of the target, not by the tile. We use it in four pieces.

Theorem 1 (Imported; reptilings, [13], as restated in [6, Thm. 4] and tabulated in [1, Table 3]). *If T is similar to R , then $N \in \{n^2\} \cup \{e^2 + f^2\} \cup \{3n^2\}$, the second family occurring exactly for right tiles and the third for the $(\pi/6, \pi/3, \pi/2)$ tile; every value in these families occurs.*

Theorem 2 (Imported; commensurable tiles, [1, Thm. 3 and Table 3]). *If all three angles of R are rational multiples of π , then the pair (T, R) and N correspond to one of the eight lines of [1, Table 3]; in particular*

$$N \in \{n^2\} \cup \{e^2 + f^2\} \cup \{2n^2\} \cup \{3n^2\} \cup \{6n^2\}.$$

Theorem 3 (Imported; non-isosceles targets, Laczkovich, [11, Thm. 4.1], as tabulated in [1, Table 4] and restated in [6, Thm. 9]). *Suppose T is not isosceles, T is not similar to R , and the angles of R are not all rational multiples of π . Then, after possibly interchanging (α, a) with (β, b) , exactly one of the following holds:*

- (a) $3\alpha + 2\beta = \pi$ and T has angles $(2\alpha, \beta, \alpha + \beta)$ or $(2\alpha, \alpha, 2\beta)$ (Group 1, shapes 1–2);
- (b) $\gamma = 2\pi/3$ and T has angles $(\alpha, 2\alpha, 3\beta)$, $(\alpha, 2\beta, 2\alpha + \beta)$, $(\alpha, \alpha + \beta, \alpha + 2\beta)$ or $(2\alpha, 2\beta, \alpha + \beta)$ (Group 2, rows II–V).

Theorem 4 (Imported; isosceles non-equilateral targets, [4, Thm. 3]). *Let T be isosceles and not equilateral, N -tiled by R , and suppose T is not similar to R . Then, possibly after renaming α and β : (i) $\gamma = 2\alpha$, or $\gamma = \pi/2$, or $\gamma = 2\pi/3$, or $3\alpha + 2\beta = \pi$; (ii) unless $\gamma = \pi/2$, the angles of R are incommensurable; (iii) the base angles of T are α , β , or $\alpha + \beta$.*

Theorem 5 (Imported; equilateral targets, Laczkovich, [12, Thm. 3.3]). *Let T be equilateral and N -tiled by R . If α is not a rational multiple of π , then $\gamma = \pi/3$ or $\gamma = 2\pi/3$. If α is a rational multiple of π , then R is equilateral, or the right triangle with a $\pi/6$ angle, or the isosceles triangle with a $\pi/6$ apex.*

The statement above is the form in which [12, Thm. 3.3] is invoked in the proof of [4, Thm. 11], which is the source we read; the 2012 paper’s own wording was not re-verified here. Note also that [12, Lemma 7.1] is the step patched in the appendix of [7].

Theorem 6 (Imported; right tiles, [3, Thm. 7.10]). *Let $N > 1$ and let R be right-angled with angles $(\alpha, \beta, \pi/2)$. Then an isosceles T is N -tileable by R only in the following cases: $N/2$ is a square (any isosceles T); $N/2 = r^2 + p^2$ (base angles β with $\tan \beta = r/p$, and this is an equivalence); N is a square (right isosceles T); $N = 6k^2$ (base angles $\pi/6$). If none of these applies, no isosceles T can be N -tiled by any right tile.*

2.2. Imported rationality and parametrization statements. Each branch needs its own licence to normalize the tile to an integer triple. There are three such licences and they are logically independent.

Theorem 7 (Imported; Beeson–Zhang rationality, [7, Thm. 1.2], quoted as [15, Thm. 1]). *Let T be tiled by R with R not similar to T , R not a right triangle, and R having incommensurable angles. Then R has commensurable sides.*

Theorem 7 is what licenses the integer normalization of the Group 2 tile recorded in (1) below. For Group 1 we do not need it: rationality there is internal to [2].

Theorem 8 (Imported; rational Group-1 tile, [2, Thm. 3]). *Let T be N -tiled by a tile with sides (a, b, c) and angles (α, β, γ) , let $3\alpha + 2\beta = \pi$, and suppose T is not similar to the tile. Then the tile is rational; specifically $s = a/c$ and b/c are rational.*

With $s = a/c = p/q$ in lowest terms, $0 < p < q$, and $b/c = 1 - s^2$ [2, Lemma 2], the tile is a positive rational multiple of $(pq, q^2 - p^2, q^2)$, and that triple is primitive: a prime dividing q^2 and pq divides q , hence divides $q^2 - p^2$ and therefore p , contrary to $\gcd(p, q) = 1$. This is the normalization used in Section 6.

Theorem 9 (Imported; the $\gamma = 2\alpha$ branch, [3, Thm. 10.5, Lemma 11.2, Lemma 11.4]). *Let T be isosceles with base angles α , $\alpha/\pi \notin \mathbb{Q}$, and let T be N -tiled by a tile with angles $(\alpha, \beta, 2\alpha)$. Then the tile is rational; after scaling, its sides are integers without a common factor and*

$$(a, b, c) = (u^2, v^2 - u^2, uv)$$

for coprime integers u, v with $u < v < 2u$; and the equal side X of T satisfies the area equation $X^2 = Nab$.

The area equation is elementary—both areas carry the factor $\sin(\alpha + \beta)$ —so only the rationality and the parametrization are genuine imports.

2.3. The two groups. Putting the pieces together, the branches that survive for a target that is neither a reptile nor covered by the commensurable or right-tile lists are the two groups

$$3\alpha + 2\beta = \pi \quad (\text{Group 1}), \quad \gamma = \frac{2\pi}{3} \quad (\text{Group 2}),$$

together with the two isolated branches $\gamma = 2\alpha$ (isosceles target) and $\gamma = \pi/3$ (equilateral target). Proposition 14 below carries out the resulting case division in full for $N = 33$. The Group 2 tile can consequently be scaled to a primitive integer triple

$$(1) \quad c^2 = a^2 + ab + b^2, \quad \gcd(a, b, c) = 1,$$

where c is opposite the 120° angle.

Lemma 10 (Arithmetic of primitive Group 2 triples). *For a primitive positive solution of (1), the integers a, b, c are pairwise coprime, $3 \nmid c$, and both c and $a - b$ are units modulo $3ab$.*

Proof. If a prime divides any two of a, b, c , the norm equation makes it divide the third, contrary to primitivity. Thus the triple is pairwise coprime. If $3 \mid c$, reduction modulo 3 gives $a \equiv b \pmod{3}$. If $3 \mid a$, then also $3 \mid b$, again impossible. Otherwise write $b = a + 3t$; then

$$a^2 + ab + b^2 = 3(a^2 + 3at + 3t^2) \equiv 3 \pmod{9},$$

whereas $c^2 \equiv 0 \pmod{9}$, a contradiction. Hence $3 \nmid c$. Finally, pairwise coprimality gives $\gcd(a - b, ab) = 1$, and $(a - b)^2 \equiv c^2 \not\equiv 0 \pmod{3}$ gives $3 \nmid (a - b)$. The stated unit claims follow. $\square$

Lemma 11 (Arithmetic of primitive 60° triples). *Let a, b, c be positive integers with*

$$c^2 = a^2 - ab + b^2, \quad \gcd(a, b, c) = 1.$$

Then a, b, c are pairwise coprime, $3 \nmid c$, $3 \nmid (a + b)$, and both c and $a + b$ are units modulo $3ab$.

Proof. If a prime ℓ divides two of a, b, c , then the norm equation in the forms

$$c^2 = a^2 - ab + b^2, \quad b^2 = c^2 - a^2 + ab, \quad a^2 = c^2 + ab - b^2$$

makes ℓ divide the square of the third, hence the third, contrary to primitivity. Thus the triple is pairwise coprime.

Next, $a^2 - ab + b^2 = (a + b)^2 - 3ab$, so

$$c^2 \equiv (a + b)^2 \pmod{3},$$

and therefore $3 \mid c$ if and only if $3 \mid (a + b)$. (Equivalently: among the nine residue pairs (a, b) modulo 3, exactly the three with $a + b \equiv 0 \pmod{3}$ make $a^2 - ab + b^2$ divisible by 3.) Suppose $3 \mid (a + b)$, hence $3 \mid c$. Then $3 \nmid a$, since $3 \mid a$ together with $3 \mid (a + b)$ would give $3 \mid b$, contradicting $\gcd(a, b) = 1$. Write $b = -a + 3k$ with $k \in \mathbb{Z}$; then

$$c^2 = a^2 - a(-a + 3k) + (-a + 3k)^2 = 3a^2 - 9ak + 9k^2.$$

As $3 \nmid a$ we have $a^2 \equiv 1 \pmod{3}$, so $c^2 \equiv 3a^2 \equiv 3 \pmod{9}$, whereas $3 \mid c$ forces $9 \mid c^2$. This contradiction gives $3 \nmid (a + b)$ and $3 \nmid c$.

Finally, pairwise coprimality gives $\gcd(c, ab) = 1$, and with $3 \nmid c$ the residue of c is a unit modulo $3ab$. Likewise $\gcd(a + b, a) = \gcd(b, a) = 1$ and $\gcd(a + b, b) = \gcd(a, b) = 1$, so $\gcd(a + b, ab) = 1$; with $3 \nmid (a + b)$ the residue of $a + b$ is a unit modulo $3ab$ as well. $\square$

The six Group 2 target shapes are listed in Table 1. We use

$$U = a + 2b, \quad V = 2a + b, \quad W = a + b.$$

Rows I–V are the five non-equilateral shapes in Laczkovich's table, with the interchange $(\alpha, a) \leftrightarrow (\beta, b)$ allowed; row E is the equilateral target.

row	target data	necessary scale condition	tile count
I	sides $k(c, c, U)$	$b \mid k$	$m^2 bU$
II	sides $k(c^2, Uc, 3bW)$	none beyond $k \in \mathbb{N}$	$3m^2 UW$
III	sides $k(ac, bV, Wc)$	none beyond $k \in \mathbb{N}$	$m^2 VW$
IV	sides $k(a, c, W)$	$b \mid k$	$m^2 bW$
V	sides $k(aU, bV, c^2)$	none beyond $k \in \mathbb{N}$	$m^2 UV$
E	equilateral, side L	$ab \mid L$	$m^2 ab$

TABLE 1. The Group 2 rays. In rows I and IV, $k = bm$; in row E, $L = abm$. In the other rows $m = k$. The arithmetic necessity is proved in Section 4; known constructions realize a tail, but generally not every multiplier.

The prime case is now unusually well cross-checked. The proof in [10] uses scale integrality and the count formulae; [9] adds a signed boundary obstruction; and [5] derives the four scalene Group 2 cases independently, delegating the other branches to the cited classification papers. We do not repeat that proof. Its role here is to close one slice of $\mathcal{S}$ and to fix the imported framework on which the composite investigation rests.

None of the prime-case results is used in any proof in this paper. Theorem 16, Proposition 14 and the ledger of Appendix E import only the classification, rationality and construction statements of Sections 2.1–2.2 and of Tables 7 and 8; the prime classification enters solely as context and as the corroborating rows of that ledger.

3. NEW EXACT TILINGS AND THE EXCLUSION OF 33

Theorem 12 (Verified exact certificates). *The following tilings exist.*

- (i) *A triangle with side lengths $(21, 55, 56)$ is tiled by 88 congruent copies of the triangle $(3, 5, 7)$.*
- (ii) *A triangle with side lengths $(36, 36, 63)$ is tiled by 189 congruent copies of the triangle $(2, 3, 4)$.*

Consequently $88m^2, 189m^2 \in \mathcal{S}$ for every $m \geq 1$.

Proof. The exact vertex coordinates are the certificates identified in Appendix A.1. For each certificate, an exact checker verifies the three squared side lengths of every tile, containment in the target, pairwise interior-disjointness by rational polygon clipping, and equality of the exact total area with the target area. By Lemma 45 these four checks force the union of the listed tiles to be exactly the reconstructed target, so that target is tiled by the listed congruent copies. Subdividing every tile quadratically into m^2 similar copies gives the final assertion. $\square$

The 88-tiling is row III of Table 1 at multiplier one: $VW = (2a + b)(a + b) = 11 \cdot 8 = 88$. Zhang’s general construction for this row, through its published uniform threshold, first produces $7128 = 88 \cdot 9^2$ tiles for $(3, 5, 7)$ [15]. Thus the exact example reduces the smallest known count on this ray by a factor 81.

The 189-tiling is Group 1 shape 5 for $(p, q) = (1, 2)$, whose ray is $21n^2$; see Section 6. The known 84- and 336-tilings give multipliers 2 and 4 [2], while the new tiling gives multiplier 3. The addition theorem below therefore yields:

Corollary 13 (The ray $21n^2$). *Let the target be the Group 1 shape-5 triangle for $(p, q) = (1, 2)$ at multiplier n , tiled by the tile $(2, 3, 4)$; at $n = 1$ this is the triangle with side lengths $(12, 12, 21)$.*

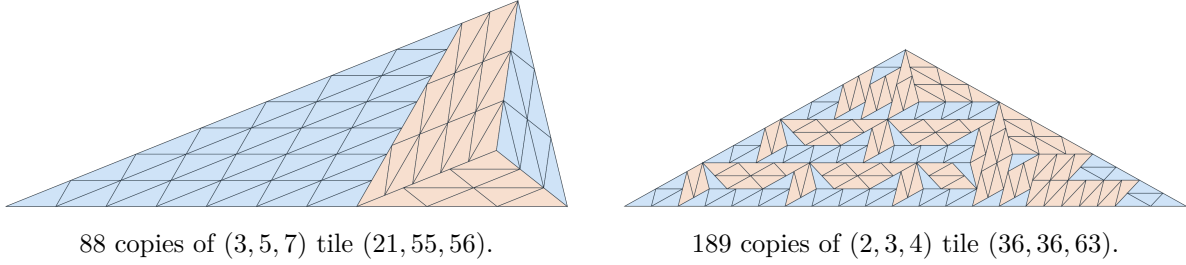


FIGURE 1. The two new tilings. These images are renderings of the exact coordinate certificates, not geometric evidence in place of them; the certificates rendered here have SHA-256 digests beginning `163327fc` ($N = 88$) and `28f33084` ($N = 189$), printed in full in Appendix A.1.

- (i) (proved, from verified certificates) *For every $n \geq 2$ a tiling exists.*
- (ii) (computer-assisted, independently certified) *For $n = 1$ no tiling exists.*

Hence a tiling on this ray exists exactly for $n \geq 2$, and $21n^2 \in \mathcal{S}$ exactly for $n \geq 2$.

Proof. (i) The known 84- and 336-tilings give the multipliers 2 and 4 [2], and Theorem 12(ii) gives the multiplier 3 as an exact coordinate certificate. The corner data of Appendix C verify the hypotheses of Proposition 27 for this shape, so the multiplier set is additively closed; since $\langle 2, 3 \rangle = \{n \geq 2\}$, every $n \geq 2$ occurs.

(ii) The exporter of Appendix A.3, run on this instance, produced the refutation certificate `N21_refutation.jsonl.gz` with 887 nodes and SHA-256 `70633336...9eaf76`, which the independent checker (SHA-256 `52d74ab3...7fd99c`) accepts. By Theorem 57 the target admits no tiling by $(2, 3, 4)$. Corroboration, not used in the proof: the primary exact polygon search exhausts the same instance in 415 states with no skipped placement, a replay with the general segment-semigroup table and boundary-invariant pruning disabled exhausts it in 674 states, both in the sense of Definition 55; and Beeson's Table 5 reports $n = 1$ as negative on the basis of a computer boundary search [2]. $\square$

Before excluding 33 we record the branch reduction in full. The point of the next proposition is that the initial list is exhaustive for a reason that can be checked line by line against named statements of named versions, rather than by a global appeal to a collection of classification papers.

Proposition 14 (Complete branch reduction for $N = 33$). *Let T be a nondegenerate Euclidean triangle tiled by 33 triangles congruent to a common tile R ; reflections are allowed and the tiling is not assumed edge-to-edge. Then, after possibly interchanging (α, a) with (β, b) and possibly reflecting, exactly one of the following mutually exhaustive cases holds.*

- (i) T is similar to R (retiling).
- (ii) T is not similar to R and all three angles of R are rational multiples of π .
- (iii) T is not similar to R , $\gamma = \pi/2$, and T is isosceles.
- (iv) $3\alpha + 2\beta = \pi$ and T has angles $(2\alpha, \beta, \alpha + \beta)$ (shape 1).
- (v) $3\alpha + 2\beta = \pi$ and T has angles $(2\alpha, \alpha, 2\beta)$ (shape 2).
- (vi) $3\alpha + 2\beta = \pi$ and T is isosceles with base angles β , i.e. $T = (\beta, \beta, 3\alpha)$ (shape 3).
- (vii) $3\alpha + 2\beta = \pi$ and T is isosceles with base angles $\alpha + \beta$, i.e. $T = (\alpha + \beta, \alpha + \beta, \alpha)$ (shape 4).
- (viii) $3\alpha + 2\beta = \pi$ and T is isosceles with base angles α , i.e. $T = (\alpha, \alpha, \alpha + 2\beta)$ (shape 5).
- (ix) $\gamma = 2\alpha$ and T is isosceles with base angles α , i.e. $T = (\alpha, \alpha, \pi - 2\alpha)$.

- (x) $\gamma = 2\pi/3$ and T is isosceles with base angles α , i.e. $T = (\alpha, \alpha, \alpha + 3\beta)$ (row I).
- (xi) $\gamma = 2\pi/3$ and $T = (\alpha, 2\alpha, 3\beta)$ (row II).
- (xii) $\gamma = 2\pi/3$ and $T = (\alpha, 2\beta, 2\alpha + \beta)$ (row III).
- (xiii) $\gamma = 2\pi/3$ and $T = (\alpha, \alpha + \beta, \alpha + 2\beta)$ (row IV).
- (xiv) $\gamma = 2\pi/3$ and $T = (2\alpha, 2\beta, \alpha + \beta)$ (row V).
- (xv) $\gamma = 2\pi/3$ and T is equilateral (row E).
- (xvi) $\gamma = \pi/3$ and T is equilateral.

In cases (iv)–(xvi) the tile is not similar to T and its angles are incommensurable, so it may be normalized to a primitive integer triple: to $(pq, q^2 - p^2, q^2)$ with $0 < p < q$, $\gcd(p, q) = 1$ in cases (iv)–(viii) by Theorem 8; to $(u^2, v^2 - u^2, uv)$ with $u < v < 2u$, $\gcd(u, v) = 1$ in case (ix) by Theorem 9; and to a solution of (1) in cases (x)–(xv) by Theorem 7. In case (xvi) the same argument applies with $c^2 = a^2 - ab + b^2$.

Tables 2 and 3 dispose of every case except (x), and case (x) leaves exactly one instance: the tile $(a, b, c) = (5, 3, 7)$ at scale $k = bm = 3$, that is, the target with sides $(21, 21, 33)$. The eliminations use only Lemma 18, Corollary 22, Proposition 38, Proposition 40, Remark 23 and the imported statements of Section 2.1.

Proof. Exhaustiveness of (i)–(xvi). If T is similar to R we are in case (i). Otherwise, if the tile angles are all rational multiples of π we are in case (ii). So assume T is not similar to R and the tile angles are incommensurable, and split on the shape of T .

If T is not isosceles, Theorem 3 gives shapes 1–2 of Group 1 and rows II–V of Group 2, i.e. cases (iv), (v) and (xi)–(xiv).

If T is isosceles and not equilateral, Theorem 4(i) leaves $\gamma \in \{2\alpha, \pi/2, 2\pi/3\}$ or $3\alpha + 2\beta = \pi$. The value $\gamma = \pi/2$ is case (iii). Part (iii) of that theorem restricts the base angles of T to α , β or $\alpha + \beta$. For $3\alpha + 2\beta = \pi$ these three options are exactly shapes 5, 3 and 4, i.e. cases (viii), (vi) and (vii). For $\gamma = 2\pi/3$ one has $\alpha + \beta = \pi/3$, so base angles $\alpha + \beta$ would make T equilateral, contrary to assumption; base angles α or β give row I, i.e. case (x), after the interchange $(\alpha, a) \leftrightarrow (\beta, b)$. For $\gamma = 2\alpha$, the pairing of target shape with tile in [1, Table 4] (Laczkovich’s second table, [11, Thm. 4.1]) is $T = (\alpha, \alpha, \pi - 2\alpha)$, i.e. base angles α ; this is case (ix).

If T is equilateral, Theorem 5 gives $\gamma \in \{\pi/3, 2\pi/3\}$, i.e. cases (xvi) and (xv).

The cases are mutually exclusive because the tile angle families $\gamma = 2\alpha$, $\gamma = \pi/2$, $\gamma = 2\pi/3$, $\gamma = \pi/3$ and $3\alpha + 2\beta = \pi$ are pairwise incompatible for a tile with incommensurable angles, and because the listed target shapes are pairwise non-similar for such a tile.

Dispositions. These are the entries of Tables 2 and 3; each is a finite arithmetic check on $N = 33$, and all of them are executable in `computation/verify_theory_report3_claims.py`. $\square$

Remark 15. Three hypotheses are load-bearing in the exhaustiveness argument and are therefore stated in the proposition rather than assumed silently: reflections are allowed, no tiling is edge-to-edge, and the case division is by the shape of T and not by the tile. The incommensurability of the tile angles is *not* a standing hypothesis of the proposition; it is the complement of case (ii). Case (iii) is the only case in which the tile angles may be commensurable while T is not similar to R (Theorem 4(ii)), which is why the right-tile branch is disposed of by Theorem 6 rather than by an arithmetic ray.

Two independent cross-checks of the Group 1 rows are available in the same source. First, 33 occurs in none of [2, Tables 1–5], and each of those tables is asserted complete to a bound exceeding 33: shape 1 to $N \leq 500$ [2, Cor. 1], shape 2 to $N \leq 2000$ [2, Table 2], shape 3 to

branch	source theorem	$N = 33$ arithmetic	disposition
(i) reptile	Thm. 1	33 is not a square; $33 = 3 \cdot 11$ with $11 \equiv 3 \pmod{4}$ to an odd power, so not $e^2 + f^2$; $33/3 = 11$ is not a square	excluded
(ii) commensurable tile	Thm. 2	$33 \notin \{n^2, e^2 + f^2, 2n^2, 3n^2, 6n^2\}$: 33 is odd, and $33/3 = 11$, $33/6 \notin \mathbb{Z}$	excluded
(iii) right tile, T isosceles	Thm. 6	33 is odd, so $N/2 \notin \mathbb{Z}$; 33 is not a square; $33 \neq 6k^2$	excluded by the last clause
(iv) G1 shape 1	[2, Thm. 4]; Cor. 37	$N = t^2(2q^2 - p^2)$ with $t \in \mathbb{N}$; $t^2 \mid 33$ forces $t = 1$, and $2q^2 - p^2 = 33$ with $p < q$ gives $q^2 < 33$, i.e. $q \leq 5$, where no p solves it	excluded: 33 is not of the form $t^2(2q^2 - p^2)$
(v) G1 shape 2	[2, Thm. 9, Table 2]; Table 4	the ray value $(2q^2 - p^2)(3q^2 - p^2)n^2$ is minimized at $(p, q) = (1, 2)$, giving $7 \cdot 11 = 77 > 33$	excluded: 33 is below the smallest shape-2 count
(vi) G1 shape 3	[2, Thm. 14, Cor. 2]; Table 4	the equation is $N(q + p)^2 = M^2(3q^2 - p^2)$, and $\gcd((q + p)^2, 3q^2 - p^2)$ forces $(q + p) \mid M$, so $N = u^2(3q^2 - p^2)$ with $M = (q + p)u$; then $u^2 \mid 33$ gives $u = 1$ and $3q^2 - p^2 = 33$, where $2q^2 < 33 < 3q^2$ leaves only $q = 4$, $p^2 = 15$	excluded: 33 is not of the form $u^2(3q^2 - p^2)$
(vii) G1 shape 4	Prop. 38; App. B	the exact ray is $N = (q^2 - p^2)m^2$; $m = 1$ and $q^2 - p^2 = 33$ give $(p, q) = (4, 7)$ and $(16, 17)$	excluded: $q \nmid m$ (Prop. 38)
(viii) G1 shape 5	Prop. 38; App. B	$N = (q^2 - p^2)(2q^2 - p^2)n^2$; $n = 1$ and the factorizations $33 = 1 \cdot 33 = 3 \cdot 11$ give $q^2 - p^2 \in \{1, 3\}$, and $(p, q) = (1, 2)$ has $2q^2 - p^2 = 7 \neq 11$	excluded: 33 is not on any shape-5 ray
(ix) $\gamma = 2\alpha$	Prop. 40	$N = (v^2 - u^2)r^2$; $r = 1$ and $v^2 - u^2 = 33$ give $(u, v) = (4, 7)$ and $(16, 17)$	excluded: $r \geq 2$ (Prop. 40)

TABLE 2. Disposition of the branches of Proposition 14 at $N = 33$, part I: preliminaries, Group 1, and $\gamma = 2\alpha$. Since $33 = 3 \cdot 11$ is squarefree, every ray equation of the form $N = Cm^2$ forces $m = 1$ and $C = 33$; this is used silently in rows (iv)–(xvi) of both parts. Rows (iv)–(vi) are stated in the strongest available form: the exclusion uses only the tiling equation of the cited theorem and needs none of that paper’s auxiliary divisibility clauses. Every row of both parts is an executable check in `computation/verify_theory_report3_claims.py`.

$N \leq 1000$ [2, Cor. 2], shape 4 to $N \leq 108$ [2, Cor. 4], and shape 5 to $N \leq 200$ [2, Cor. 5].¹ Second, a direct sweep of the five tiling equations of [2, Table 6] over $s = p/q$ with $q \leq 400$

¹The range of the last of these is printed as $N \leq 200$ in the statement of [2, Cor. 5] and as $N \leq 500$ in the caption of that paper’s Table 5, and the corollary is labelled “(to Theorem 14)” where Theorem 19 is meant. Both bounds exceed 33, so the discrepancy is immaterial here.

branch	source theorem	$N = 33$ arithmetic	disposition
(x) G2 row I	Cor. 22	$N = m^2bU$ with $U = a + 2b$; $bU = 33$ has the unique primitive solution $(a, b, c) = (5, 3, 7)$, $U = 11$, at $m = 1$, hence $k = bm = 3$	survives: row I, $(5, 3, 7)$, $k = 3$, target $(21, 21, 33)$
(xi) G2 row II	Cor. 22	$N = 3m^2UW$ with $U = a + 2b$, $W = a + b$; the least value of UW over primitive triples is $11 \cdot 8 = 88$, at $(a, b) = (5, 3)$, so $3UW \geq 264 > 33$	excluded: 33 is below the smallest row-II count
(xii) G2 row III	Cor. 22	$N = m^2VW$ with $V = 2a + b$; the least value of VW over primitive triples is $11 \cdot 8 = 88 > 33$, at $(a, b) = (3, 5)$	excluded: 33 is below the smallest row-III count
(xiii) G2 row IV	Cor. 22	$N = m^2bW$; the least value of bW is $3 \cdot 8 = 24$, so $m = 1$ and $bW = 33$; the factorizations $(b, W) = (1, 33), (3, 11)$ give $a = W - b = 32, 8$ and $c^2 = 1057, 97$, neither a square	excluded: no primitive triple has $bW = 33$
(xiv) G2 row V	Cor. 22	$N = m^2UV$; the least value of UV over primitive triples is $11 \cdot 13 = 143 > 33$	excluded: 33 is below the smallest row-V count
(xv) G2 row E	Cor. 22 ($ab \mid L$)	$N = m^2ab$; $m = 1$ and $ab = 33$ give $(a, b) = (1, 33), (3, 11)$ with $c^2 = 1123, 163$, neither a square	excluded: no primitive triple has $ab = 33$
(xvi) $\gamma = \pi/3$, T equilateral	Rem. 23	$N = m^2ab$ with $c^2 = a^2 - ab + b^2$; $m = 1$ and $ab = 33$ give $c^2 = 1057, 97$, neither a square	excluded: no primitive 60° triple has $ab = 33$

TABLE 3. Disposition of the branches of Proposition 14 at $N = 33$, part II: Group 2 and the 60° equilateral branch.

shows that shapes 1, 2, 3 and 5 admit no solution at all with $N = 33$, and that shape 4 admits only $(p, q, M) \in \{(4, 7, 3), (16, 17, 1), (4, 29, 5), (29, 37, 2), (17, 49, 4)\}$, all of which fail the exact shape-4 ray $N = (q^2 - p^2)m^2$ except the first two, which fail $q \mid m$. This sweep is section H of the executable audit `computation/verify_theory_report3_claims.py`, which also validates the transcribed equation set against all 53 printed rows of [2, Tables 1–5].

Theorem 16 (Computer-assisted, independently certified). $33 \notin \mathcal{S}$.

Proof. By Proposition 14, a hypothetical 33-tiling reduces to exactly one target: the isosceles triangle $(21, 21, 33)$ tiled by $(5, 3, 7)$, which is row I of Table 1 at $k = 3$; this agrees with Beeson’s effective isosceles enumeration [3].

For that instance the exporter of Appendix A.3 produced the refutation certificate `N33_refutation.jsonl.gz` with 23157 nodes and SHA-256 `613c52dd...c21919`. The independent checker (SHA-256 `52d74ab3...7fd99c`) accepts it in under one minute. The checker was written from the certificate specification alone and shares no code with the search; the export run disables memoization and canonicalization, so the certificate is a tree of regions in absolute

coordinates, and it uses only the three reduced pruning rules, so it depends on neither Theorem 19 nor the general form of Lemma 53. By Theorem 57 the target admits no tiling by $(5, 3, 7)$.

Two earlier exhaustions of the same instance corroborate this and are not used in the proof: the primary exact polygon search exhausts it in 6479 states with all pruning enabled, and a replay with the general segment-semigroup table and the boundary-invariant pruning disabled returns EXHAUSTED in 22850 states. Both runs are exhausted in the sense of Definition 55, and both come from the same implementation. $\square$

Remark 17. The boundary invariant of Section 4 does not exclude this candidate: it satisfies the Laurent boundary congruence and even admits the required exact- N signed inventory. Theorem 16 is therefore a small but concrete demonstration that any final solution needs interior order, incidence, or connectivity information.

4. A LAURENT BOUNDARY QUOTIENT

Lemma 18 (Boundary integrality). *If a convex polygon is tiled by a triangle with integer side lengths, every side of the polygon has integer length and is a concatenation of complete tile edges.*

Proof. Let P be the convex polygon, S one of its sides, and ℓ the supporting line of S , so that P lies in one closed half-plane bounded by ℓ and

$$P \cap \ell = S,$$

because the face of P cut out by the supporting line of a side is that side. Let a tile Δ meet the relative interior of S in a nondegenerate segment. Then ℓ supports Δ as well, so $\Delta \cap \ell$ is a face of Δ ; it has positive dimension, hence is a complete edge of Δ . Moreover $\Delta \cap \ell \subseteq P \cap \ell = S$, so that edge cannot extend beyond either endpoint of S . Distinct such edges have disjoint relative interiors. Every point of S lies in some tile; a tile that does not contribute one of the edges just described meets ℓ in at most one point, and there are finitely many tiles, so those edges cover S up to a finite set. Their union is closed, hence equals S , and their integer lengths add to the length of S . No edge-to-edge hypothesis is used. $\square$

In a Group 2 tiling, every directed edge direction has a unique expression

$$\theta = j\pi/3 + m\alpha, \quad j \in \mathbb{Z}/6, \quad m \in \mathbb{Z},$$

after an anchor is chosen; here α/π is irrational [7]. Put $d = a - b$ and define the character

$$\chi(j, m) = (-1)^j z^m \in \mathbb{Z}[z, z^{-1}]^\times.$$

The pair (j, m) is determined by θ . Indeed, if $j\pi/3 + m\alpha \equiv j'\pi/3 + m'\alpha \pmod{2\pi}$, then $(m - m')\alpha \in \pi\mathbb{Q}$, so $m = m'$ because α/π is irrational, and then $(j - j')\pi/3 \in 2\pi\mathbb{Z}$ gives $j \equiv j' \pmod{6}$. The irrationality used here is proved in Appendix D: $2\cos\alpha = U/c$ is rational and lies strictly between 1 and 2, so it is not a rational integer, whereas $\alpha/\pi \in \mathbb{Q}$ would force it to be one; see also [7]. We therefore write $\chi(\theta)$ for $\chi(j, m)$.

For a directed segment PQ of direction θ , set $B(PQ) = |PQ|\chi(\theta)$, initially in $\mathbb{R}[z, z^{-1}]$, and extend B additively to oriented polygonal chains. By Lemma 18, complete tile and target boundaries have integral coefficients.

Theorem 19 (Laurent boundary quotient). *Let a triangle be tiled by a primitive Group 2 tile satisfying (1), and write $R = \mathbb{Z}[z, z^{-1}]$ and $\pi_I : R \rightarrow R/I$ for the quotient map. Then*

$$(2) \quad B(\partial T) \in I = (d + cz, d + cz^{-1}) \subseteq R,$$

and $3ab \in I$. Moreover the following two maps are mutually inverse ring isomorphisms:

$$(3) \quad \begin{aligned} \bar{\varepsilon} : R/I &\longrightarrow \mathbb{Z}/(3ab), \quad \text{induced by } \varepsilon : R \rightarrow \mathbb{Z}/(3ab), \quad z \mapsto -dc^{-1}, \\ \bar{\iota} : \mathbb{Z}/(3ab) &\longrightarrow R/I, \quad \text{induced by } \mathbb{Z} \hookrightarrow R \xrightarrow{\pi_I} R/I. \end{aligned}$$

Here ε is well defined because c and d are units modulo $3ab$ (Lemma 10), and $\bar{\iota}$ is well defined because $I \cap \mathbb{Z} = (3ab)$. Consequently (2) is equivalent to the single evaluation

$$(4) \quad B(\partial T)|_{z=-dc^{-1}} \equiv 0 \pmod{3ab}.$$

Proof. The counterclockwise boundary of a direct tile, with its a -edge anchored, has value

$$d + cz^{-1};$$

a reflected tile has value $d + cz$, up to multiplication by a Laurent unit. Subdivide every tile edge at every tiling vertex. In this common refinement, each elementary interior segment is incident with exactly one tile on each side, so it occurs twice with opposite orientations and cancels. This proves (2); no edge-to-edge hypothesis is being imposed.

Since $c^2 - d^2 = 3ab$,

$$(5) \quad c(d + cz) - dz(d + cz^{-1}) = (c^2 - d^2)z = 3abz,$$

and z is a unit of R , so $3ab \in I$.

By Lemma 10 the residues of c and d are units modulo $3ab$, so $-dc^{-1}$ is a unit there and ε is a well-defined ring homomorphism. It kills both generators of I :

$$\varepsilon(d + cz) = d - cdc^{-1} = 0, \quad \varepsilon(d + cz^{-1}) = d - c^2d^{-1} = -3abd^{-1} = 0.$$

Hence $I \subseteq \ker \varepsilon$ and ε induces $\bar{\varepsilon}$ on R/I .

For the other direction, work in R/I . By (5) we have $3ab = 0$ there, and $\gcd(c, 3ab) = 1$ gives integers c', n with $cc' = 1 + 3abn$, so c' inverts c in R/I . The two generators of I therefore give

$$z = -c'd, \quad z^{-1} = -c'd$$

in R/I , so every Laurent monomial is congruent there to an integer and $\mathbb{Z} \hookrightarrow R \xrightarrow{\pi_I} R/I$ is surjective. Its kernel is $I \cap \mathbb{Z}$, and

$$(3ab) \subseteq I \cap \mathbb{Z} \subseteq \ker \varepsilon \cap \mathbb{Z} = (3ab),$$

the first inclusion by (5) and the last because ε restricted to $\mathbb{Z}$ is reduction modulo $3ab$. Hence $I \cap \mathbb{Z} = (3ab)$ and $\bar{\iota}$ is a well-defined ring isomorphism. Finally $\bar{\varepsilon} \circ \bar{\iota}$ is the identity of $\mathbb{Z}/(3ab)$, again because $\varepsilon|_{\mathbb{Z}}$ is reduction modulo $3ab$; as both maps are bijections, $\bar{\iota}$ and $\bar{\varepsilon}$ are mutually inverse. This proves (3), and $\ker \varepsilon = I$ gives the equivalence of (2) with (4). $\square$

Corollary 20 (Universal property of the boundary quotient). *Let A be a commutative ring and let $\phi : \mathbb{Z}[z, z^{-1}] \rightarrow A$ be a ring homomorphism with*

$$\phi(d + cz) = \phi(d + cz^{-1}) = 0.$$

Then there is a unique ring homomorphism $\psi : \mathbb{Z}/(3ab) \rightarrow A$ with $\phi = \psi \circ \varepsilon$. In particular $3ab \cdot 1_A = 0$, the element $\phi(c)$ is invertible in A , and $\phi(z) = -\phi(d)\phi(c)^{-1}$. Consequently, for every tiling as in Theorem 19,

$$\phi(B(\partial T)) = \psi\left(B(\partial T)|_{z=-dc^{-1}}\right),$$

so every boundary invariant obtained by specializing the character χ along such a ϕ factors through the cyclic quotient (3), and no such specialization can obstruct a candidate that satisfies (4).

Proof. The hypotheses give $I \subseteq \ker \phi$, and $\ker \varepsilon = I$ by Theorem 19, so ϕ factors as $\psi \circ \varepsilon$; the factor ψ is unique because ε is surjective. Then $3ab \cdot 1_A = \phi(3ab) = \psi(0) = 0$, and $\phi(c) = \psi(\varepsilon(c))$ is invertible because $\varepsilon(c)$ is a unit of $\mathbb{Z}/(3ab)$ by Lemma 10, whence $\phi(z) = -\phi(c)^{-1}\phi(d)$ from $\phi(d + cz) = 0$. Applying ψ to $\varepsilon(B(\partial T))$ gives the displayed identity, and the last assertion is that identity read backwards: if (4) holds, the argument of ψ vanishes. $\square$

Remark 21 (Scope of Corollary 20). Signed characters, root-of-unity specializations, and reductions modulo a divisor of $3ab$ all satisfy the hypothesis of Corollary 20 and are therefore already contained in (4). The corollary says nothing about invariants that are not ring specializations of χ : bounds on the number or on the signs of the N admissible tile terms representing $B(\partial T)$, and any use of edge order, incidence, or connectivity, lie outside its scope. This is what Theorem 16 exhibits concretely.

Corollary 22 (Exact Group 2 arithmetic). *The six necessary conditions and count rays in Table 1 hold for every Group 2 tiling.*

Proof. Evaluate the counterclockwise target boundary in (4). Up to multiplication by units modulo $3ab$, the six residues are

row	I	II	III	IV	V	E
$B(\partial T)$	$3a^2k$	0	0	$3ak$	0	$3L$.

Since $\gcd(a, b) = 1$, the nonzero entries give $b \mid k$ in rows I and IV and $ab \mid L$ in row E. The area equations give the counts in Table 1. The primitive target side vectors make the remaining scale integral; detailed row polynomials and gcd checks are recorded in Appendix B. $\square$

Remark 23 (What the invariant closes). The quotient proves Zhang’s divisibility conjecture for equilateral 120° tiles, including the nonsquarefree cases not covered by his lemma. It also removes the congruence hypothesis from Zhang’s row-V count formula [15]. The same proof works for 60° tiles after replacing $c^2 = a^2 + ab + b^2$ by $c^2 = a^2 - ab + b^2$ and $d = a - b$ by $d = a + b$: the one-tile polynomials become, up to units, $d - cz^{-1}$ and $d - cz$, while $c^2 - d^2 = -3ab$, so that

$$c(d - cz) + dz(d - cz^{-1}) = (d^2 - c^2)z = 3abz,$$

and Lemma 11 supplies exactly the units the argument of Theorem 19 consumes: c and $d = a + b$ are invertible modulo $3ab$. The evaluation is $z \mapsto dc^{-1}$ and the quotient is again $\mathbb{Z}/(3ab)$. Conversely, by Corollary 20 all ideal-membership obstructions obtained by specializing this Laurent character are exhausted; the invariants excluded from that statement are listed in Remark 21.

5. CONSTRUCTION THRESHOLDS AND ADDITIVE MULTIPLIER SETS

For $x, L > 0$, an *ideal trapezoid* $T(x, L)$ has parallel sides x and $x + L$, base angles $\pi/3$, and two legs of length L .² Its Laurent boundary is $3L$, independent of x , so Theorem 19 gives the necessary condition $ab \mid L$.

The construction below uses two statements of [15] verbatim, both read with y meaning the leg, as that paper’s own proofs do.

Lemma 24 (Imported; basic ideal trapezoid, [15, Lemma 1]). *The ideal trapezoid with short parallel side $x = a^2 + b^2$ and leg $L = ab$ is tiled by (a, b, c) ; it splits into the three triangles similar to (a, b, c) with integer factors a, b, c , hence into $a^2 + b^2 + c^2$ tiles.*

²Zhang [15] interchanges the leg and long-side symbols; here L is the leg.

Lemma 25 (Imported; parallelograms, [15, Lemma 2]). *A parallelogram whose angles are, in order, $2\pi/3$, $\pi/3$, $2\pi/3$, $\pi/3$, and whose side lengths are x and ab , is tiled by (a, b, c) whenever $x > ab - a - b$. The proof glues two tiles along c into an (a, b, a, b) parallelogram, forms the (ab, a, ab, a) and (ab, b, ab, b) parallelograms from it in the two orientations, and concatenates them; it therefore proves the sharper statement that the parallelogram is tiled whenever $x \in a\mathbb{N}_0 + b\mathbb{N}_0$.*

The sharper reading of Lemma 25 is what Lemma 26 uses; the Frobenius bound $x > ab - a - b$ of [15, Prop. 2] is only the crude sufficient condition for $x \in a\mathbb{N}_0 + b\mathbb{N}_0$.

Lemma 26 (Trapezoid construction). *Let $R = (a, b, c)$ be a primitive 120° tile. If*

$$x, L \in \mathbb{N}, \quad ab \mid L, \quad x - (a^2 + b^2) \in a\mathbb{N}_0 + b\mathbb{N}_0,$$

then $T(x, L)$ is tiled by

$$\frac{L(2x + L)}{ab}$$

copies of R .

Proof. Lemma 24 supplies the basic trapezoid $T(a^2 + b^2, ab)$ and Lemma 25 the $\pi/3$ -parallelograms of side ab and base a or b . Write $L = kab$ and slice $T(x, L)$ into k layers of leg ab . In layer i , remove a copy of the basic trapezoid. The remainder is a parallelogram whose base is

$$x + (i - 1)ab - (a^2 + b^2) \in a\mathbb{N}_0 + b\mathbb{N}_0,$$

so it is tiled by concatenating the two parallelograms of Lemma 25. The area ratio gives the stated number of tiles. $\square$

For $x = rab$, the semigroup condition in Lemma 26 is equivalent to

$$r \geq R(a, b) := \left\lceil \frac{a}{b} \right\rceil + \left\lceil \frac{b}{a} \right\rceil.$$

Indeed, if $rab - a^2 - b^2 = Aa + Bb$, reduction modulo b and a gives $A = ub - a$ and $B = va - b$ for integers $u \geq \lceil a/b \rceil$, $v \geq \lceil b/a \rceil$, and then $r = u + v$. The converse follows from the same formulae. In the cyclic equilateral construction of Theorem 28 below, which assembles an equilateral triangle of side $(r + s + t)ab$ from three ideal trapezoids with parameters r, s, t , choose the parameters $R, R, m - 2R$; Lemma 26 tiles each of the three trapezoids because all parameters are at least R , exactly when $m \geq 3R$. This gives an equilateral tiling for every such multiplier. At $(a, b, c) = (32, 45, 67)$ this gives the smaller multiplier $m = 9$.³ The count 116640 is already classical, so this changes the construction theory rather than $\mathcal{S}$ itself.

Proposition 27 (Multiplier addition). *Let Δ be a triangle and R a tile. Suppose one vertex of Δ has adjacent sides L_1, L_2 , while two adjacent sides e_1, e_2 of R meet at that angle or its supplement, and $L_1/e_1, L_2/e_2 \in \mathbb{N}$. If $r, s \in \mathbb{N}$ and $r\Delta$ and $s\Delta$ are tiled by R , then $(r + s)\Delta$ is tiled by R .*

Proof. Write $\Delta = \text{conv}(O, A, B)$. In $(r + s)\Delta$, put $P = rA$, $Q = rB$, and $D = rA + sB$. The three regions

$$\triangle OPQ, \quad P + s\Delta, \quad \text{conv}(P, Q, (r + s)B, D)$$

have disjoint interiors and fill $(r + s)\Delta$; the last is a parallelogram with side lengths $r|B - A|$ and $s|B|$. Labeling the chosen corner of Δ as B makes these rL_1 and sL_2 . Subdivide the parallelogram into an $r(L_1/e_1)$ by $s(L_2/e_2)$ grid of parallelograms with sides e_1, e_2 , and split every cell along the diagonal determined by whether the included angle is the tile angle or its

³The threshold printed in [15, Conjecture 2] is 12, so that conjectured threshold is not sharp.

supplement. The two halves are congruent copies of R by SAS. No edge-to-edge hypothesis is used. $\square$

For a fixed ray, let $\mathcal{A}$ be the set of positive multipliers that admit a tiling. Here a *numerical semigroup* means a cofinite additive submonoid of $\mathbb{N}_0$. The hypotheses of Proposition 27 hold on all six Group 2 rows and on Group 1 shapes 4 and 5; the integer ratios are given in Appendix C.

The transfer of an equilateral tail to the other rows is imported. Call L *equiconstructible* for (a, b, c) if the equilateral triangle of side L is tiled by (a, b, c) .

Theorem 28 (Imported; equilateral construction, [15, Thm. 4]). *Put $M_Z = 3[(c^2 - a - b)/(ab)]$. Then mab is equiconstructible for every integer $m \geq M_Z$, and hence an m^2ab -tiling exists for every such m . The proof assembles an equilateral triangle of side $(r + s + t)ab$ from three ideal trapezoids with parameters $r, s, t \geq M_Z$.*

Proposition 29 (Imported; row transfers from an equilateral core, [15, Props. 8–10]). *Suppose mab is equiconstructible for (a, b, c) . Then, in the notation of Table 1,*

- (a) [15, Prop. 8(1)] *the triangle $(mbc, mbc, mb(a + 2b))$, of shape row I with $k = bm$, is tiled by $m^2b(a + 2b)$ copies of (a, b, c) ;*
- (b) [15, Prop. 9(1)] *the triangle $(mab, mbc, mb(a + b))$, of shape row IV with $k = bm$, is tiled by $m^2b(a + b)$ copies;*
- (c) [15, Prop. 10(1)] *the triangle $(mac, (b + 2a)mb, (a + b)cm)$, of shape row III with $k = m$, is tiled by $m^2(b + 2a)(a + b)$ copies.*

In each case the statement with a and b interchanged also holds.

Proposition 30 (Imported; row-V to row-II transfer, [15, Prop. 11]). *If the triangle $((a + 2b)am, (b + 2a)bm, c^2m)$ —row V at multiplier m —is tiled by (a, b, c) , then the triangle $(c^2m, (a + 2b)mc, 3(a + b)mb)$ —row II at multiplier m —is tiled by $3m^2(a + 2b)(a + b)$ copies of (a, b, c) , and likewise with a and b interchanged.*

Propositions 29 and 30 are used only as implications: (a)–(c) consume only equiconstructibility of mab , supplied here by the tail $m \geq 3R(a, b)$ proved above, and Proposition 30 consumes the row-V interval produced by the bisector dissection of Appendix C.

Corollary 31 (Finite generator windows). *For every primitive Group 2 tile and every row of Table 1, $\mathcal{A} \cup \{0\}$ is a numerical semigroup. More precisely:*

- (i) *in rows E, I, III, and IV, every $m \geq 3R(a, b)$ occurs and*

$$\mathcal{A} \cup \{0\} = \langle \mathcal{A} \cap [1, 6R(a, b) - 1] \rangle;$$

- (ii) *in rows V and II, every $s \geq \lceil 3R(a, b)/2 \rceil$ occurs and*

$$\mathcal{A} \cup \{0\} = \langle \mathcal{A} \cap [1, 2\lceil 3R(a, b)/2 \rceil - 1] \rangle.$$

Thus each Group 2 ray is determined by finitely many exact tiling questions.

Proof. Additive closure is Proposition 27. The equilateral tail $m \geq 3R(a, b)$ was proved immediately above from Lemma 26 and the cyclic construction of Theorem 28. Proposition 29(a)–(c) transfers that tail to rows I, IV, and III. For rows V and II, a bisector dissection transfers an equilateral tiling of side $2sab$ to row V at multiplier s ; the dissection and count identity are in Appendix C. Proposition 30 transfers the same interval to row II. Finally, if an additively closed set contains every integer at least C , then each $n \geq 2C$ is $C + (n - C)$ and induction shows that the set is generated by its members below $2C$. The complement is finite, so $\mathcal{A} \cup \{0\}$ is a numerical semigroup. $\square$

Theorem 32 (Trapezoid tilings for $(3, 5, 7)$: the constructive half). *For the tile $(3, 5, 7)$ and $L = 15$, the ideal trapezoid $T(x, 15)$ is tiled by $(3, 5, 7)$ for every integer $x \in \{29, 31, 32\}$ and every integer $x \geq 34$.*

Proof. Here $ab = 15 = L$, so Lemma 26 applies whenever $x - 34 \in 3\mathbb{N}_0 + 5\mathbb{N}_0$, that is for $x = 34, 37, 39, 40$ and every $x \geq 42$. The seven exact coordinate certificates listed in Appendix A.6 cover

$$x = 29, 31, 32, 35, 36, 38, 41.$$

Each passes the checks of Appendix A.1, including the trapezoid-shape assertions, so by Lemma 45 its tiles cover the intended trapezoid exactly. $\square$

Theorem 33 (Trapezoid tilings for $(3, 5, 7)$: the exclusion half; computer-assisted). *For the tile $(3, 5, 7)$, $L = 15$, and every integer x with $1 \leq x \leq 33$ and $x \notin \{29, 31, 32\}$, the ideal trapezoid $T(x, 15)$ admits no tiling by $(3, 5, 7)$.*

Proof. The boundary inventory excludes every positive $x \notin \langle 3, 5, 7 \rangle$, namely $x \in \{1, 2, 4\}$. For the remaining values the exact searcher was run; the search found precisely the 27 exhausted cases

$$\langle 3, 5, 7 \rangle \cap [1, 33] \setminus \{29, 31, 32\},$$

each terminating with status exhausted in the sense of Definition 55, with no skipped placement and no unsupported geometric case. Completeness of each search tree is Lemmas 47, 48, 49, 51 and 52, and every prune used is one of the necessary conditions of Lemma 54. The campaign comprises 34 searches and 5,699,235 states; Appendix A.6 gives the case table and the completeness scope. $\square$

This half depends on a single exhaustion implementation: no refutation certificate has been exported for these instances. The pipeline of Appendix A.3 applies to trapezoid targets without change, and the deferred campaign, with the commands it needs, is recorded with the certificate specification in the ancillary files.

Corollary 34 (The one-leg family at $L = 15$). *For the tile $(3, 5, 7)$, $L = 15$, and every integer $x \geq 1$,*

$$T(x, 15) \text{ is tileable} \iff x \in \{29, 31, 32\} \cup \{x \in \mathbb{N} : x \geq 34\}.$$

The implication “ $\Leftarrow$ ” is Theorem 32 and is proved. The implication “ $\Rightarrow$ ” is Theorem 33 and therefore carries that theorem’s evidence class: computer-assisted, one implementation, no independent certificate.

Corollary 34 gives the complete local picture for this one-leg family, with its two evidence classes separated by Theorems 32 and 33.⁴ The necessary divisibility $ab \mid L$ remains valid.

6. THE REMAINING ARITHMETIC BRANCHES

For Group 1, normalize $a/c = p/q$ with $0 < p < q$ and $\gcd(p, q) = 1$. A primitive tile has sides

$$(pq, q^2 - p^2, q^2).$$

The five target shapes lie on the rays in Table 4. Shapes 1–3 are characterized by Beeson [2]. Shapes 4–5 are necessity statements; their sufficiency remains open in general.

The shape-1 count is used below, so we state it in the present normalization. Beeson’s shape-1 analysis is carried by two statements.

⁴The literal inequality in [15, Conjecture 5] does not hold as printed and already conflicts with the basic construction in [15, Lemma 1].

Theorem 35 (Imported; shape-1 necessity, [2, Thm. 4]). *Let $3\alpha + 2\beta = \pi$ and let T have angles $(2\alpha, \beta, \alpha + \beta)$ and be N -tiled by a tile with angles α and β . Then $M^2 + N = 2K^2$ for some positive integers M, K with $M^2 < N$, where $s = 2\sin(\alpha/2) = M/K$; and N is a square times a product of distinct primes of the form $8n \pm 1$.*

Theorem 36 (Imported; shape-1 divisibility, [2, Thm. 6]). *Let T have angles $(2\alpha, \beta, \alpha + \beta)$, let integers N, K, M solve $N = 2K^2 - M^2$, and suppose there is an N -tiling of T by a tile with angles (α, β, γ) and sides (a, b, c) with $a/c = M/K$. Then $K \mid M^2$.*

Corollary 37 (Shape-1 ray). *With the normalization $a/c = p/q$ of this section, the count of every shape-1 tiling is*

$$N = q^2(2q^2 - p^2)n^2, \quad n \in \mathbb{N},$$

and conversely every such N occurs ([2, Thm. 5]).

Proof. By Lemma 2 of [2], $s = 2\sin(\alpha/2) = a/c = p/q$. Let (M, K) be as in Theorem 35. From $M/K = p/q$ with $\gcd(p, q) = 1$ we get $M = pt$, $K = qt$ for an integer $t \geq 1$, whence $N = 2K^2 - M^2 = t^2(2q^2 - p^2)$. Theorem 36 applies to this (M, K) and gives $qt \mid p^2t^2$, i.e. $q \mid p^2t$, i.e. $q \mid t$. Writing $t = qn$ gives the displayed ray. Conversely, for $(M, K) = (pqn, q^2n)$ one has $M^2 + N = 2q^4n^2 = 2K^2$, $M^2 < N$ and $K \mid N$, so [2, Thm. 5] produces an N -tiling, by the tile $n \cdot (pq, q^2 - p^2, q^2)$. $\square$

Only Corollary 37 is used later, in the proof of Proposition 38. We do not use the printed statement [2, Thm. 7], which asserts the existence of an N -tiling if and only if the tiling equation has a solution with $K \nmid M^2$; that inequality contradicts Theorems 5 and 6 of the same paper and the proof of Theorem 7 itself, and is a typographical negation.

Proposition 38. *For every Group 1 shape-4 tiling, the natural multiplier m satisfies $q \mid m$; hence its count has the shape-4 form in Table 4. Every shape-5 tiling has the shape-5 count in that table, without an additional parity factor. The multiplier sets for shapes 4 and 5, with zero adjoined, are additive submonoids of $\mathbb{N}_0$.*

Proof. For shape 4, an exact target insertion gives equal side $X = (q^2 - p^2)qm$. Attach a qm -scaled tile along that side. The union is a shape-1 target with count $(2q^2 - p^2)m^2$. Corollary 37 says its count is $q^2(2q^2 - p^2)n^2$, forcing $m = qn$. For shape 5, cancellation in the Group 1 boundary ideal forces the target scale C to be divisible by $q^2 - p^2$, giving the listed count. The factorization and target insertion are recorded in Appendix B. Additive closure is Proposition 27. $\square$

Unlike Corollary 31, this proposition supplies neither a tail nor an a priori finite generator window for shapes 4 and 5.

shape	target angles	necessary count ray
1	$(2\alpha, \beta, \alpha + \beta)$	$q^2(2q^2 - p^2)n^2$
2	$(2\alpha, \alpha, 2\beta)$	$(2q^2 - p^2)(3q^2 - p^2)n^2$
3	$(\beta, \beta, 3\alpha)$	$q^2(3q^2 - p^2)n^2$
4	$(\alpha + \beta, \alpha + \beta, \alpha)$	$q^2(q^2 - p^2)n^2$
5	$(\alpha, \alpha, \alpha + 2\beta)$	$(q^2 - p^2)(2q^2 - p^2)n^2$

TABLE 4. Group 1 rays. In shape 4 the natural scale m is divisible by q , so $m = qn$; in shape 5 there is no additional parity factor, including when p and q are both odd.

Remark 39 (Erratum: the clause $g \mid M$ of [2, Thm. 19]). Proposition 38 deliberately does not use a further divisibility clause carried by the published shape-5 necessity theorem. We record why, because a reader who imports that clause will exclude values that are not excluded.

The cited statement. [2, Thm. 19] reads, verbatim: “Suppose $3\alpha + 2\beta = \pi$. Let ABC be isosceles with base angles α . Suppose ABC is N -tiled by the tile whose integer sides (a, b, c) have no common factor, and whose angles are (α, β, γ) . Let $s = a/c$ and $g = \gcd(a, c)$, and let M be the coloring number of the tiling. Then g divides M , $M^2 \leq 2N$, and (M, s) is a solution of the ‘isosceles- α tiling equation’

$$N = M^2 \frac{(1+s)(2-s^2)}{(1-s)(2+s)^2}.$$

Translation of notation. In the normalization of this section the tile is $(a, b, c) = (pq, q^2 - p^2, q^2)$, so

$$s = \frac{a}{c} = \frac{p}{q}, \quad g = \gcd(pq, q^2) = q.$$

Beeson’s M is the coloring number of the tiling, not our multiplier n . Substituting $s = p/q$ in the displayed equation gives

$$\frac{N}{M^2} = \frac{(q+p)(2q^2 - p^2)}{(q-p)(2q+p)^2},$$

and combining this with the shape-5 ray $N = (q^2 - p^2)(2q^2 - p^2)n^2$ of Proposition 38 yields

$$M = (q-p)(2q+p)n.$$

Hence, on the ray, the printed clause is equivalent to a statement about our own multiplier:

$$g \mid M \iff q \mid (q-p)(2q+p)n \iff q \mid p^2n \iff q \mid n,$$

the last step by $\gcd(p, q) = 1$.

A counterexample. Take $(p, q) = (1, 2)$, tile $(2, 3, 4)$, and $n = 3$. Then $s = 1/2$, $g = 2$, and

$$\frac{N}{M^2} = \frac{(1 + \frac{1}{2})(2 - \frac{1}{4})}{(1 - \frac{1}{2})(2 + \frac{1}{2})^2} = \frac{\frac{3}{2} \cdot \frac{7}{4}}{\frac{1}{2} \cdot \frac{25}{4}} = \frac{21}{25},$$

so $M = 15$ and $N = 189$. The coloring number is not merely predicted by the equation: the same paper’s coloring relation $M = \mu(1-s)(2+s)$ with $\mu = X/b$ applied to the certified tiling of Theorem 12(ii), which has $X = 36$ and $b = 3$, gives $\mu = 12$ and $M = 12 \cdot \frac{1}{2} \cdot \frac{5}{2} = 15$; and [2, Table 5] itself lists the row $N = 189$, $M = 15$, $(a, b, c) = (2, 3, 4)$. But $g = 2$ does not divide $M = 15$. Since Theorem 12(ii) exhibits an exact 189-tiling of $(36, 36, 63)$ by $(2, 3, 4)$, the printed divisibility clause fails on a realized tiling.

Extent. Of the 15 rows of [2, Table 5], 12 violate $g \mid M$; the three that satisfy it are the two rows marked “yes” ($N = 84$, $M = 10$ and $N = 336$, $M = 20$) and the row $N = 306$, $M = 21$, $(3, 8, 9)$. A condition satisfied exactly on the realized rows of a table generated *without* it has the shape of a sufficiency-side condition that has migrated into a necessity statement; compare the isosceles- β case, where $g \mid M$ is precisely the matching sufficient condition [2, Thm. 16]. We flag this reading and do not resolve it. Two textual observations point at the same place: the printed proof of Theorem 19 concludes from $g \mid X^2$ that $g \mid X$ “since g is squarefree” by [2, Lemma 8], whose squarefreeness clause fails for $g = 4$ —the tiles $(12, 7, 16)$ and $(4, 15, 16)$ of that paper’s own Tables 1, 2 and 5 have $\gcd(a, c) = 4$ —and the same proof asserts that $bc^3(a+c)$ is not divisible by g^5 , whereas the first clause of [2, Lemma 8] gives $c = g^2$ and hence $c^3 = g^6$. The proof also contains an unresolved cross-reference, printed as “By Lemma ??”.

Independence. Nothing in this paper depends on the clause. The equation half of [2, Thm. 19] is consistent with all of our data and is used only as a cross-check; the shape-5 ray of Proposition 38 is proved from the Group-1 boundary ideal in Appendix B, and the shape-4 half from Corollary 37. Had the clause been imported it would have “excluded” the shape-5 values 161, 280 and 465, none of which is excluded, and it would have contradicted Theorem 12(ii).

The remaining isosceles branch has tile angle $\gamma = 2\alpha$ [3]. With coprime integers $u < v < 2u$, its primitive side triple is

$$(a, b, c) = (u^2, v^2 - u^2, uv).$$

Proposition 40 ($\gamma = 2\alpha$ ray). *Every tiling of a target $(\alpha, \alpha, \pi - 2\alpha)$ in this branch, with α/π irrational, has*

$$N = (v^2 - u^2)r^2$$

for an integer $r \geq 2$. (By Theorem 4 and the pairing of target shape with tile in [1, Table 4], this is the only target shape the branch admits; compare case (ix) of Proposition 14.)

Proof. The branch carries its own Laurent boundary quotient. The target boundary lies in the ideal of $\mathbb{Z}[z, z^{-1}]$ generated by the two one-tile boundary polynomials, and cancelling their common quadratic factor in the Laurent domain forces the scale t of the equal side $X = ut$ into the residual ideal, whose intersection with $\mathbb{Z}$ is $(v^2 - u^2)$. This is an ideal membership, not merely a rationality statement: it gives $t = (v^2 - u^2)r$ with r an integer, and inserting $X = u(v^2 - u^2)r$ into the area equation $X^2 = Nu^2(v^2 - u^2)$ of Theorem 9 gives the stated square ray. At $r = 1$, the target boundary is too short to carry the necessary tile edges, giving $r \geq 2$. The full boundary insertion and capacity argument are given in Appendix B. $\square$

Remark 41. A stronger assertion $r \geq 3$ is part of the conditional forced-layer program, not Proposition 40. This distinction affects several small values and is made explicit in Appendix F.

7. FULL-SIDE TRANSFERS AND ORDERED SEGMENT DATA

A *one-patch row transfer* adjoins a quadratically tiled t -scaled copy of the tile along one complete target side, where $t \in \mathbb{N}$, and straightens one seam endpoint. The following result closes this particular construction mechanism; it does not classify partial-side or multi-patch attachments.

Write $X(a, b; m)$ for the Group 2 target in row $X \in \{E, I, II, III, IV, V\}$ of Table 1, with normalized multiplier m , and write $S_i(n)$ for the Group 1 target in shape i of Table 4, with its displayed normalized multiplier n .

Proposition 42 (One-patch row-transfer calculus). *Under the standing incommensurable-angle hypotheses, the complete row-to-row lists, up to reflection and $(a, \alpha) \leftrightarrow (b, \beta)$, are as follows:*

$$\begin{aligned} \text{Group 2: } & E(a, b; m) \longrightarrow IV(a, b; m), \\ & IV(a, b; m) \longrightarrow I(a, b; m), \quad III(b, a; m), \\ & V(a, b; m) \longrightarrow II(a, b; m), \\ & IV(a, b; as) \longrightarrow IV(a, b; cs), \\ & V(a, b; as) \longrightarrow III(a, b; cs), \\ & V(a, b; bs) \longrightarrow III(b, a; cs); \end{aligned}$$

and

$$\begin{aligned} \text{Group 1: } S_4(n) &\longrightarrow S_1(n), S_5(n), \\ S_1(n) &\longrightarrow S_2(n), S_3(n), \\ S_1(bs) &\longrightarrow S_1(cs), \\ S_4(ps) &\longrightarrow S_4(qs). \end{aligned}$$

No Group 2 row-to-row transfer of this kind lands in row V , and no Group 1 transfer of this kind starts from S_5 .

Proof. The finite angle enumeration, attachment scales, merged sides, and count identities are given in Appendix C. That enumeration also contains the uninformative cases in which the union is itself a scaled copy of the tile; these are reptilings, not transfers among the named non-similar target rows. $\square$

The transfer graph explains two persistent asymmetries. A row- V target has no incoming one-patch edge, while shape 5 has no outgoing one-patch edge. Consequently those two branches require a multi-patch, partial-side, or interior-seam construction.

Proposition 43 (Relation lattice and pure-edge words). *Let (a, b, c) be a primitive positive solution of $c^2 = a^2 + ab + b^2$. After possibly interchanging a and b , there are coprime integers $r > s > 0$, with $r \not\equiv s \pmod{3}$, such that*

$$a = r^2 - s^2, \quad b = 2rs + s^2, \quad c = r^2 + rs + s^2.$$

Put

$$R_c = (r + s, s, -r), \quad R_b = (s, -r, s).$$

Then:

- (i) R_c, R_b form a $\mathbb{Z}$ -basis of

$$\{(x, y, z) \in \mathbb{Z}^3 : ax + by + cz = 0\}.$$

Thus every signed equality of unordered full-edge inventories along a collinear segment has two unique integer charges.

- (ii) Orient a chain of consecutive a -edges from left to right and suppose all supported tiles lie in the same half-plane. With $\rho = e^{i\pi/3}$, call the third vertex over $[x, x + a]$

$$A : x + a + b\rho, \quad B : x + b\rho^2.$$

Every support word has the form $B^j A^{n-j}$ and hence at most one defect. The analogous assertion holds for a chain of b -edges.

- (iii) At the possible $B \rightarrow A$ defect, the two exposed third vertices and the shared endpoint bound a wedge with sides

$$(c, c, 2a + b).$$

Its area divided by the tile area is $(2a + b)/a$, which is never an integer.

Proof. See Appendix D. $\square$

The proposition is deliberately one-dimensional. The two exposed c -edges at a defect are seeds, not yet proved paths, and the nonintegral wedge is not a detachable row-I macrotile. A complete seam theorem would still have to prove continuation through every encountered vertex, embeddedness and noncrossing of suitable outermost paths, pairing or boundary termination, geometric realization of the relation charges by strips, successive exposure of peelable layers, and termination with finitely describable interfaces. A finite inventory basis alone does not imply any of these statements.

Remark 44 (Herdt’s 720-tiling). The orientation-changing parallelogram construction exhibited by Beeson [3, Fig. 22] is made explicit in Appendix C. A vector-figure audit of Herdt’s (4, 5, 6) construction in [3, Fig. 24] finds ten quadratic patches and three orientation-changing blocks, each split into two uniform subparallelograms; their count inventory sums to 720. This is a check of the displayed macro inventory, not an exact coordinate certificate for the figure.

8. CONSEQUENCES AND THE REMAINING GEOMETRIC GAP

The principal conceptual consequence is negative but useful: after Theorem 19, further specializations of the same $\chi(j, m) = (-1)^j z^m$ ideal-membership invariant will not settle the surviving multipliers. Promising next objects must retain ordering or incidence, or impose exact-term constraints absent from ideal membership. Three such structures are already available:

- (1) orient every c -edge of a Group 2 tile from its α endpoint to its β endpoint. The resulting flow is conserved at every non-corner vertex and forces directed c -edge paths between charged target corners; such a path may run along the target boundary;
- (2) row III at multiplier one is exactly a two-grid staircase problem: half of a $(2a + b) \times (a + b)$ parallelogram is cut along a legal c -edge diagonal. The 88-tiling is the first worked instance.
- (3) every unordered full-edge segment mismatch has the two unique charges of Proposition 43, while a pure a - or b -edge support word has at most one ordered defect.

These facts do not yet exclude or construct a new multiplier, but they name finite, ordered geometric objects that a final argument can classify. Their short proofs are included in Appendix D.

9. CONCLUSION

The results above change the shape of the full problem in four ways. First, the small exact tilings at 88 and 189 show that published construction thresholds can be far from sharp. Second, the Laurent quotient settles the χ -based ideal-membership arithmetic, sharply identifying what that boundary method can and cannot decide. Third, multiplier addition turns every Group 2 ray into a numerical semigroup determined by an explicit finite window. Fourth, the complete one-patch transfer graph and the relation/word normal forms isolate the next genuinely geometric object: an ordered seam whose continuation, pairing, and interfaces must be controlled in the presence of T-junctions. Thus the Group 2 branch reduces to explicit finite per-ray windows; the other branches still require comparable construction or obstruction theorems before such a finite reduction is available for the full Erdős Problem 634.

The immediate mathematical targets are small and concrete: decide the first unknown multipliers on the row-I, row-IV, row-V, Group 1 shape-5, and $\gamma = 2\alpha$ rays; close or replace the forced-layer induction; prove a geometrically realizable seam calculus rather than only its inventory charges; and extend the search-independent refutation certificates of Appendix A.3 to the remaining negative computations. The manuscript is structured so that those results can be added without changing the distinction between proved and conditional claims.

APPENDIX A. EXACT CERTIFICATES AND POSITIVE VERIFICATION

Data and code availability. The accompanying arXiv-ready source package contains an `anc` directory with the verification manifest, exact certificate data, the primary $N = 33$ search and two stored exhaustion outputs, the corrected separately written diagnostic searcher and its regression witness, and the arithmetic audit programs and captured outputs. Computational

paths displayed below are relative to the root of that ancillary tree. The manifest records exact commands, environments, qualifications, and SHA-256 digests.

A.1. Certificate semantics. Coordinates are pairs (x, w) representing $(x, w\sqrt{D})$, with rational x, w . Here $D = 3$ for Group 2 and $D = 15$ for the displayed Group 1 (2, 3, 4) instance. The verifier reads only the tile triangles from a JSON certificate; it does not read the search tree. It reconstructs their convex hull and, when supplied with the theorem's expected data, asserts the number of hull corners and the exact multiset of target side lengths. For the seven trapezoid certificates it further checks the cyclic order of the four side lengths, parallelism of the two unequal bases, and both diagonal lengths; thus the asserted hull is the intended ideal trapezoid $T(x, 15)$, not merely a quadrilateral with the same side multiset. It checks:

- (1) each tile has the required three squared side lengths;
- (2) every tile is contained in the reconstructed convex target;
- (3) every pair of tiles has intersection of exact area zero; and
- (4) the sum of the exact tile areas equals the exact target area.

The step from these four checks to literal coverage of the target is the following elementary lemma, used again by the refutation checker of Appendix A.3. Call a closed set $X \subseteq \mathbb{R}^2$ a *body* if $X = \overline{\text{int } X}$; closed polygons, closed triangles, and closures of components of open polygonal sets are bodies.

Lemma 45 (Coverage). *Let T be a bounded body and let $P_1, \dots, P_k \subseteq T$ be bodies with pairwise disjoint interiors and $\sum_{i=1}^k \text{area}(P_i) = \text{area}(T)$. Then $\bigcup_{i=1}^k P_i = T$.*

Proof. For $i \neq j$, disjointness of the interiors gives $P_i \cap P_j \subseteq \partial P_i$, a Lebesgue-null set, so $\text{area}(\bigcup_i P_i) = \sum_i \text{area}(P_i) = \text{area}(T)$. Put $U = \bigcup_i P_i$, a closed subset of T . Then $\text{int } T \setminus U$ is open and null, hence empty, so $\text{int } T \subseteq U$ and therefore $T = \overline{\text{int } T} \subseteq U \subseteq T$. $\square$

Lemma 46 (Containment by area). *Let X be a body and Y a closed set. Then $X \subseteq Y$ if and only if $\text{area}(X \cap Y) = \text{area}(X)$.*

Proof. One direction is trivial. Conversely, if some $x \in \text{int } X \setminus Y$ then, Y being closed, a disk about x lies in $X \setminus Y$, so $\text{area}(X \setminus Y) > 0$. Hence $\text{int } X \subseteq Y$ and $X = \overline{\text{int } X} \subseteq Y$. $\square$

Check (1) makes every listed triangle a congruent copy of the tile; check (2) is containment in the reconstructed target; check (3) forces pairwise disjoint interiors, since two closed triangles with a common interior point share a disk and hence have intersection of positive area; and check (4) is the area identity. Lemma 45 then gives literal coverage. Lemma 46 is what licences deciding containment by an exact intersection area, as both the positive verifier and the refutation checker do.

certificate	tiles	chirality split	pairs checked	SHA-256
$N = 88$	88	58/30	3828	163327fc06cf8f98...aefbbd
$N = 189$	189	77/112	17766	28f33084443ba303...d5096

TABLE 5. Exact-certificate audit.

The full SHA-256 digests are

$N = 88$: 163327fc06cf8f98639b38743e8d796c9fdcd86d722485229a8f047e94aefbbd
 $N = 189$: 28f33084443ba303e6758511e616c3f9bbf3de1657e3422eee6680118a0d5096.

The files are:

```
computation/tiling_search/results/N88_tiling.json
computation/tiling_search/results/N189_tiling.json
computation/tiling_search/verify_tiling.py.
```

The SVG and plain-text coordinate exports beside the JSON files are convenience views of the same data.

A.2. The exact exhaustion method. The primary search selects a deterministic convex corner P of each remaining polygonal component.

Lemma 47 (Convex-corner branching). *Let a polygonal region be tiled by finitely many non-degenerate triangles, and let P be a convex vertex of one remaining component. Fix either incident boundary ray at P . The tile adjacent to that ray has a vertex at P and a complete tile edge beginning at P along the ray. Therefore branching over all tile angles, both assignments of the adjacent side lengths, and both chiralities is exhaustive.*

Proof. Intersect a sufficiently small disk about P with the component. It is a wedge of angle less than π , partitioned by the sectors of the incident tiles. The point P cannot lie in a tile interior, and it cannot lie in the relative interior of a tile edge, since the wedge contains no complete line through P . Thus every incident tile has a vertex at P . The tile sector adjacent to the chosen boundary ray has one of its sides on that ray, and that side is a complete edge of the tile. The edge may continue beyond a later reflex boundary vertex; exact containment, not the length of the first boundary segment, decides whether the placement is legal. $\square$

Lemma 47 bounds the branching to a fixed finite list, which both the search and the certificate format use verbatim. Throughout, $\theta_1, \theta_2, \theta_3$ are the tile angles, θ_i opposite the side s_i , and $\text{Rot}(\varphi)$ is rotation by φ . In both branches $\cos \theta_i$ is rational by the law of cosines and $\sin \theta_i = w_i \sqrt{D}$ with $w_i \in \mathbb{Q}$: in Group 2, $\cos \alpha = (a+2b)/2c$ and $\sin \alpha = (a/2c)\sqrt{3}$; in Group 1, $\cos \alpha = (2q^2 - p^2)/2q^2$ and $\sin \alpha = (p/2q^2)\sqrt{D}$ with $D = 4q^2 - p^2$. So every candidate below has coordinates in $\mathbb{Q}(\sqrt{D})$ as soon as the chosen boundary edge has rational length, which the searcher and the checker both test.

Lemma 48 (The flush placement list). *Let the polygonal region R be positively oriented and tiled by congruent copies of the tile, let P be a convex boundary vertex of R with interior angle $\theta_P < \pi$, and let u be the unit direction of one of the two boundary rays at P . Let Δ be a tile of the tiling that has a vertex at P and a complete edge starting at P along u , as supplied by Lemma 47. Then:*

- (i) $\Delta = (P, P + \ell u, P + \ell' \text{Rot}(\varepsilon \theta_i) u)$ for some $i \in \{1, 2, 3\}$, some ordering (ℓ, ℓ') of the two sides adjacent to θ_i , and some $\varepsilon \in \{+1, -1\}$. Hence Δ occurs in the twelve-element list indexed by i ascending, then the two orderings, then $\varepsilon = +1$ before $\varepsilon = -1$;
- (ii) $\theta_i \leq \theta_P$, equivalently $\cos \theta_i \geq \cos \theta_P$;
- (iii) $\varepsilon = +1$ if u points from P to the next vertex in the positive orientation, and $\varepsilon = -1$ if it points to the previous one. The six candidates carrying the other sign are contained in no region with P and u as described.

Consequently the at most six candidates obtained from (i) with the sign of (iii) and with $\theta_i \leq \theta_P$ form a complete list of the tiles that can be flush at P along u .

Proof. By Lemma 47, Δ has a vertex at P , and the side of Δ on the chosen ray is a complete tile edge. Let θ_i be the angle of Δ at P ; its two sides at P are the two sides adjacent to θ_i , one of them along u and the other along a rotation of u through θ_i in one of the two senses. Side-angle-side then determines Δ , which is (i).

Choose $\epsilon > 0$ so small that $B = B(P, \epsilon)$ meets ∂R only in the two boundary edges at P . Then $R \cap B$ is the wedge W of angle θ_P swept counterclockwise from the direction u_+ of the next vertex to the direction u_- of the previous one, and $\Delta \cap B$ is the sector of angle θ_i at P . Since $\Delta \subseteq R$, that sector lies in W , so $\theta_i \leq \theta_P$; both angles lie in $(0, \pi)$, where the cosine is strictly decreasing, giving the stated inequality. This is (ii).

For (iii), $\theta_P < \pi$ makes W a subset of the closed half-plane to the left of the line $\mathbb{R}u_+$, so W contains no direction obtained from u_+ by a clockwise rotation through an angle in $(0, \pi)$. A candidate with $u = u_+$ and $\varepsilon = -1$ has its third vertex in such a direction, so the points of the segment from P to that vertex which lie in B are not in R , and the candidate is not contained in R . The case $u = u_-$ is the mirror image. $\square$

The program enumerates exactly the at most six candidates of Lemma 48: it selects the deterministic convex corner P and the ray towards the next vertex, discards a tile angle with $\cos \theta_i < \cos \theta_P$, and rotates counterclockwise, which is $\varepsilon = +1$. The certificate format of Appendix A.3 instead records all twelve entries of Lemma 48(i) and leaves both the angle bound and the containment decision to the checker.

Lemma 49 (Necessary conditions for a placement). *Let R be a simple positively oriented polygonal region tiled by congruent copies of the tile and let Δ be one of the tiles, so $\Delta \subseteq R$ and $\text{int } \Delta \subseteq \text{int } R$. Then:*

- (a) *no edge of Δ and edge of R cross transversally at a point interior to both;*
- (b) *no vertex of R lies in the interior of Δ ;*
- (c) *the three vertices of Δ lie in R ;*
- (d) *the centroid of Δ lies in the interior of R ;*
- (e) *for every $V \in \partial \Delta \cap \partial R$, the interior cone of Δ at V is contained in the interior cone of R at V ; in particular the two extreme directions of the tile cone and their sum lie in the closed interior cone of R at V .*

Proof. Since $\text{int } \Delta$ is open and contained in R , it is contained in $\text{int } R$.

(a) Suppose an edge of Δ crosses a boundary edge e of R transversally at x , interior to both. As R is simple, x lies in the relative interior of no other boundary edge, so a small disk meets R in the closed half-disk on the left of e . The tile edge has points arbitrarily close to x strictly on the right of e ; these are not in R , contradicting $\Delta \subseteq R$.

(b) A vertex v of R with $v \in \text{int } \Delta$ would satisfy $v \in \text{int } R$, contradicting $v \in \partial R$.

(c) is $\Delta \subseteq R$, and (d) holds because the centroid lies in $\text{int } \Delta \subseteq \text{int } R$.

(e) Choose $\epsilon > 0$ with $B = B(V, \epsilon)$ meeting ∂R only in the boundary edges through V and $\partial \Delta$ only in the tile edges through V . Then $R \cap B$ and $\Delta \cap B$ are the sectors of the interior cones C_R and C_Δ at V , of angle $< \pi$ if V is a vertex of the respective polygon and $= \pi$ if V lies in the relative interior of one of its edges. Now $\Delta \subseteq R$ gives $C_\Delta \subseteq C_R$. The two extreme directions of C_Δ are limits of directions of $\text{int } C_\Delta$, and their sum is a direction of $\text{int } C_\Delta$; all three lie in the closed cone C_R . $\square$

Conditions (a)–(e) are, in this order, the five clauses of the program’s `fit_test`. Clause (e) is tested at the three tile vertices and at every vertex of R lying on an edge of Δ . Any further point of $\partial \Delta \cap \partial R$ lies in the relative interior of an edge of each; the two edges are then collinear by (a), and either the two cones coincide, so that (e) is vacuous, or they are opposite half-planes, which $\Delta \subseteq R$ already excludes. Only necessity is used: the search rejects a placement when a listed condition fails, and no claim is made that the list is sufficient for containment. A placement that passes it but is not contained is caught at the next step, by the area identity of Lemma 51, and is reported as a skipped placement (Definition 55).

Definition 50 (Pieces). The *pieces* of a closed set $S \subseteq \mathbb{R}^2$ are the closures of the connected components of $\text{int } S$.

Lemma 51 (Exact boundary subtraction). *Let R be a closed simple polygon, positively oriented, and let $\Delta \subseteq R$ be a nondegenerate closed triangle such that $\partial R \cap \partial \Delta$ contains a nondegenerate segment; this is the situation of every flush placement of Lemma 47. Put $S = \overline{R \setminus \Delta}$ and perform:*

- (S1) *split every edge of ∂R at every vertex of Δ lying in its relative interior and every edge of $\partial \Delta$ at every vertex of R lying in its relative interior, orienting ∂R positively and $\partial \Delta$ negatively; call the resulting directed segments half-edges;*
- (S2) *delete every pair of half-edges with the same endpoints and opposite directions;*
- (S3) *at each vertex v , let the successor $\sigma(h)$ of a half-edge h ending at v be the half-edge issuing from v whose direction comes first in clockwise order starting from the reverse of the direction of h ;*
- (S4) *discard every closed σ -walk enclosing zero area, after merging collinear runs of vertices.*

Then:

- (a) *after (S2) every surviving half-edge occurs exactly once, has S locally on its left and the closure of the complement of S locally on its right, and ∂S is the union of the surviving half-edges;*
- (b) *σ is a permutation of the surviving half-edges, and its corners $(h, \sigma(h))$ correspond bijectively to the local sectors of S at the vertices;*
- (c) *the cycles of σ are exactly the positively oriented boundary walks of the pieces of S , one cycle per piece, and each is a simple closed polygon after merging collinear runs;*
- (d) *every cycle encloses positive area, so (S4) discards nothing.*

Consequently the pieces of S are simple polygons with pairwise disjoint interiors whose union is S , their areas sum to $\text{area}(R) - \text{area}(\Delta)$, and $\bigcup_i \partial P_i \subseteq \partial S \subseteq \partial R \cup \partial \Delta$.

Proof. Elementary segments. Two half-edges from the same boundary meet only at endpoints, because R and Δ are simple and (S1) only subdivides. Suppose x lies in the relative interiors of a half-edge $e \subseteq \partial R$ and a half-edge $f \subseteq \partial \Delta$. If e and f were not collinear they would cross transversally, and f would have points arbitrarily near x outside the closed half-plane that contains R locally, contradicting $\Delta \subseteq R$. So they are collinear. The endpoints of f are vertices of Δ or vertices of R , and by (S1) and simplicity neither can lie in the relative interior of e ; symmetrically for the endpoints of e . Hence e and f have the same endpoints. Thus distinct undirected half-edges have disjoint relative interiors.

Cancellation and sides. Each undirected segment occurs at most once in ∂R and at most once in $\partial \Delta$, both being simple closed curves. Suppose it occurs in both, and let x be a relative interior point. Locally R is the closed half-disk on the left of the ∂R direction and Δ the closed half-disk on the left of the $\partial \Delta$ direction; $\Delta \subseteq R$ forces the two directions to agree, so the pair cancels in (S2), and locally $R \setminus \Delta = \emptyset$, whence a neighbourhood of x misses S . If the segment occurs only in ∂R , then by the previous paragraph its relative interior misses $\partial \Delta$, and it misses $\text{int } \Delta \subseteq \text{int } R$; so locally $S = R$ is the closed left half-disk and the complement of R the open right one. If it occurs only in $\partial \Delta$, its relative interior lies in $\text{int } R$, and locally Δ is the closed half-disk left of the $\partial \Delta$ direction while S is the other closed half-disk, which is on the left of the reversed direction used in (S1). This proves the side statement in (a), and it shows that every surviving half-edge lies in ∂S while no cancelled segment does. Since $\partial S \subseteq \partial R \cup \partial \Delta$, the union of the surviving half-edges is ∂S . Finally, after (S1) both boundary walks are closed, so in-degree equals out-degree at every vertex, and (S2) removes one incoming and one outgoing half-edge at each endpoint of a cancelled pair; the surviving chain is balanced.

Local sectors. Fix a vertex v and $\epsilon > 0$ so small that $B = B(v, \epsilon)$ meets the surviving half-edges only in radial segments at v . By (a), $\partial S \cap B$ is the union of those radii, so $S \cap B$ is a finite union of closed circular sectors with pairwise disjoint interiors, each bounded by two of the radii and containing no other radius. Distinct half-edges at v span distinct rays: two with the same ray would either coincide, which (S2) excludes, or contain each other's endpoint in a relative interior, which (S1) excludes. “ S on the left” says that the material of S along the ray of an outgoing half-edge lies on its counterclockwise side, and along the ray of an incoming half-edge on its clockwise side. Hence for each sector the clockwise-most bounding ray belongs to an outgoing half-edge and the counterclockwise-most to an incoming one; in particular no two sectors are adjacent, incoming and outgoing rays alternate, and turning clockwise from an incoming ray the first ray met is the outgoing ray of the same sector. That is exactly the rule (S3), so σ maps the half-edges ending at v bijectively to those issuing from v , and corners correspond to sectors. Being a bijection at every vertex, σ is a permutation, which is (b).

The complement of S is connected. By hypothesis $\partial R \cap \partial \Delta$ contains a nondegenerate segment, hence a cancelled half-edge e . At a relative interior point of e a small ball misses S , and it splits into the part inside $\text{int } \Delta$ and the part outside R . Therefore $\text{int } \Delta \cup \text{relint}(e) \cup (\mathbb{R}^2 \setminus R)$ is a connected subset of the complement of S . As $\mathbb{R}^2 \setminus S = (\mathbb{R}^2 \setminus R) \cup (\Delta \setminus \partial S)$ and every point of $\Delta \setminus \partial S$ lies in $\text{int } \Delta$ or in the relative interior of a cancelled half-edge, the complement $K = \mathbb{R}^2 \setminus S$ is connected and unbounded.

One sector per piece per vertex. Suppose two distinct sectors at a vertex v belonged to the same piece P . Join an interior point of one to an interior point of the other by an arc in $\text{int } P$ and close it through v : this gives a Jordan curve $J \subseteq P \subseteq S$ which, near v , consists of two arcs through v . These two arcs separate a small disk at v into two parts, and each contains one of the two complementary gaps between the two sectors; one gap therefore lies inside J and the other outside. Both consist of points of K , and K also contains points far away, outside J . So K meets both complementary regions of J while being connected and disjoint from J — a contradiction. In particular no σ -cycle visits a vertex twice, so every cycle is a simple closed polygon.

Cycles and pieces. For a surviving half-edge h , the material of S on the left of h near its relative interior lies in a unique piece $P(h)$. The sector of the corner $(h, \sigma(h))$ lies in that piece and is adjacent along $\sigma(h)$ to the material on the left of $\sigma(h)$, so $P(\sigma(h)) = P(h)$: the map P is constant on cycles. Conversely $P(h) = P$ holds precisely when $h \subseteq \partial P$. Every cycle contains a half-edge of ∂R : the surviving half-edges of $\partial \Delta$ form proper subarcs of the simple closed curve $\partial \Delta$, since at least one of its segments cancelled, and subarcs contain no cycle. Let J be a cycle of ∂P and $e \subseteq J$ a half-edge of ∂R . The complement of R is connected and unbounded, and $J \subseteq S \subseteq R$, so the bounded complementary region of J is contained in R ; the points just to the right of e are outside R and hence in the unbounded complementary region of J , while the material of P lies just to the left. Since $\text{int } P$ is connected and disjoint from J , it follows that $\text{int } P$ is contained in the bounded region enclosed by J , and that J is traversed counterclockwise, proving (d). If a second cycle $J' \subseteq \partial P$ existed, then $J' \subseteq \partial P \subseteq \text{int } \bar{P}$ and $J' \cap J = \emptyset$ would put J' inside J ; but J' also contains a half-edge of ∂R , whose right-hand side consists of points outside R lying in a neighbourhood contained in the bounded region enclosed by J , hence in R — a contradiction. So each piece carries exactly one cycle and each cycle bounds exactly one piece, which is (c).

The closing statements follow: pieces are simple polygons by (c), have pairwise disjoint interiors and union S by Definition 50 (because $S = \overline{\text{int } S}$, as $\text{int } R \setminus \Delta$ is open and dense in S), their areas sum to $\text{area}(S) = \text{area}(R) - \text{area}(\Delta)$, and $\partial P_i \subseteq \partial S$ because a point of $\text{int } S$ has a ball inside a single component. $\square$

Subtraction in `computation/tiling_search/searcher2.py` is exactly the steps (S1)–(S4): all component and tile edges are split at all incidences, half-edges of net opposite multiplicity cancel, the surviving half-edges are traced by the successor rule (S3), and walks of zero area after merging collinear runs are dropped. A flush tile edge spanning several collinear boundary segments therefore needs no special case; by Lemma 51(c) it can split off further pieces, as in the program’s regression with two such pieces. Three of these steps are guarded. Each stored piece is required to be a simple polygon, which Lemma 51(c) guarantees and the program re-verifies under its `--paranoid` flag; a negative traced area, which Lemma 51(d) excludes, is likewise rejected; and the area identity $\text{area}(R) = \text{area}(\Delta) + \sum_i \text{area}(P_i)$ is checked at every placement. A failure of any of the three is recorded as a skipped placement, so by Definition 55 the run is inconclusive and supports no nonexistence claim. All orientation, containment, area, and intersection predicates are rational in $\mathbb{Q}(\sqrt{D})$.

Lemma 52 (Components). (i) *Let S be a closed polygonal region with pieces $P_1, \dots, P_k$. Then S is tileable by congruent copies of a given triangle if and only if every P_i is.*
(ii) *Let R be tiled by triangles $\Delta_1, \dots, \Delta_n$, put $\Delta = \Delta_1$, and let $C_1, \dots, C_k$ be bodies with pairwise disjoint interiors, each interior disjoint from $\text{int } \Delta$, such that $\Delta \cup C_1 \cup \dots \cup C_k = R$ and $\bigcup_j \partial C_j \subseteq \partial R \cup \partial \Delta$. Then every Δ_i with $i \geq 2$ is contained in exactly one C_j , and for every j the tiles contained in C_j tile C_j .*

Proof. (ii) Fix $i \geq 2$. Then $\text{int } \Delta_i \subseteq \text{int } R$, so $\text{int } \Delta_i \cap \partial R = \emptyset$. Also $\text{int } \Delta_i \cap \partial \Delta = \emptyset$: a point of that intersection would have a disk inside $\text{int } \Delta_i$ meeting $\text{int } \Delta$, because $\Delta = \overline{\text{int } \Delta}$, contradicting interior disjointness of the tiles. Hence $\text{int } \Delta_i$ misses $\bigcup_j \partial C_j$ and misses Δ . As $\text{int } \Delta_i \subseteq R = \Delta \cup \bigcup_j C_j$, we get $\text{int } \Delta_i \subseteq \bigcup_j \text{int } C_j$, a disjoint union of open sets; $\text{int } \Delta_i$ is connected, so it lies in a single $\text{int } C_j$, and $\Delta_i = \overline{\text{int } \Delta_i} \subseteq C_j$. Uniqueness is clear since the interiors are disjoint.

Let I_j index the tiles contained in C_j . Their interiors are disjoint and contained in C_j , so $\sum_{i \in I_j} \text{area}(\Delta_i) \leq \text{area}(C_j)$. The sets $\Delta, C_1, \dots, C_k$ have pairwise disjoint interiors and union R , so $\text{area}(R) = \text{area}(\Delta) + \sum_j \text{area}(C_j)$, while $\sum_{i \geq 2} \text{area}(\Delta_i) = \text{area}(R) - \text{area}(\Delta)$. Since every $i \geq 2$ lies in exactly one I_j , summing the inequalities forces equality for each j , and Lemma 45 then shows that the tiles indexed by I_j cover C_j . They therefore tile it.

(i) The pieces are bodies with pairwise disjoint interiors and union S , and $\bigcup_i \partial P_i \subseteq \partial S$; so the argument just given, with Δ absent, shows that every tile of a tiling of S lies in exactly one piece and that the tiles inside a piece tile it. Conversely, the union of tilings of the pieces has pairwise disjoint interiors and covers S , hence is a tiling of S . $\square$

Together with Lemma 51, whose pieces satisfy the hypotheses of (ii) with $C_j = P_j$, this is the induction step of the search: after a placement, each tile of a hypothetical completion belongs to exactly one piece, the pieces are independent subproblems, and one untileable piece kills the placement.

Lemma 53 (Convex-end segment pruning). *If a polygonal region tiled by triangles with side lengths a, b, c has a boundary segment whose two endpoints are convex, then its length belongs to $a\mathbb{N}_0 + b\mathbb{N}_0 + c\mathbb{N}_0$.*

Proof. A tile meeting the relative interior of the segment in a nondegenerate interval has a complete edge on the same supporting line. Such a tile edge cannot extend past either endpoint: extension beyond a convex endpoint leaves the local interior wedge. These complete tile edges partition the segment, so their lengths add to its length. $\square$

Lemma 54 (The pruning conditions are necessary). *Let the closed polygonal region R be tiled by n congruent copies of a tile with sides a, b, c and angles $\theta_1, \theta_2, \theta_3$, and put $\theta_{\min} = \min_i \theta_i$. Then:*

- (i) (area budget) $\text{area}(R)/\text{area}(\text{tile}) = n$, a positive integer;
- (ii) (wedge) the interior angle at every convex boundary vertex of R is a sum of at least one and at most $\lfloor \pi/\theta_{\min} \rfloor$ tile angles; in particular it is at least $\theta_{\min}$;
- (iii) (short edge) a boundary edge of R whose two endpoints are convex has length in $a\mathbb{N}_0 + b\mathbb{N}_0 + c\mathbb{N}_0$ and, being positive, at least $\min(a, b, c)$;
- (iv) (Laurent and character bounds, Group 2) with the notation of Section 4, $B(\partial R)$ has integer coefficients, lies in $I = (d + cz, d + cz^{-1})$, and satisfies $B(\partial R)|_{z=1} = M(a - b + c)$ and $B(\partial R)|_{z=-1} = M'(a - b - c)$ with integers M, M' such that $|M|, |M'| \leq n$ and $M \equiv M' \equiv n \pmod{2}$;
- (v) (child budget) if Δ is a tile of the tiling and some piece of $\overline{R \setminus \Delta}$ has area which is not a positive integer multiple of the tile area, then no tiling of R contains Δ ; in particular such a placement may be discarded.

Proof. (i) The tiles have pairwise disjoint interiors and union R , so their areas add.

(ii) Let P be a convex boundary vertex with interior angle $\theta_P < \pi$, and intersect a small disk about P with R , obtaining the wedge of angle θ_P . A tile containing P cannot contain it in its interior, since $P \in \partial R$, and cannot contain it in the relative interior of an edge, since that edge would put a full diameter through P inside R , which a wedge of angle less than π does not contain. So every tile at P has a vertex there, the corresponding sectors partition the wedge, and θ_P is the sum of their angles: a nonempty sum of tile angles, each at least $\theta_{\min}$, hence with at most $\lfloor \theta_P/\theta_{\min} \rfloor \leq \lfloor \pi/\theta_{\min} \rfloor$ summands.

(iii) This is Lemma 53 together with the observation that the partition of a segment of positive length into complete tile edges is nonempty.

(iv) Subdivide every tile edge at every vertex of the tiling of R . Each elementary interior segment is incident with one tile on each side, so it occurs twice with opposite orientations and cancels, exactly as in the proof of Theorem 19. Hence

$$B(\partial R) = \sum_{j=1}^n \pm z^{k_j} (d + cz^{\varepsilon_j}), \quad \varepsilon_j \in \{1, -1\}, \quad k_j \in \mathbb{Z},$$

which has integer coefficients and lies in I . Every Laurent unit evaluates to ± 1 at $z = 1$ and at $z = -1$, while $d + cz^{\pm 1}$ evaluates to $d + c = a - b + c$ at $z = 1$ and to $d - c = a - b - c$ at $z = -1$. So $B(\partial R)|_{z=1}$ is a sum of n signs times $a - b + c$, and similarly at $z = -1$; the bound and the parity follow.

(v) By Lemma 51 the pieces of $\overline{R \setminus \Delta}$ satisfy the hypotheses of Lemma 52(ii), so a tiling of R containing Δ tiles every piece, and (i) applies to that piece. $\square$

Every prune the program applies is one of the conditions of Lemma 54, in the following correspondence. The area budget is (i), applied both to a component and, before recursion, to each of its pieces, which is (v). The wedge prune rejects a component with a convex vertex whose interior angle is smaller than every tile angle, a special case of (ii). The selected-edge prune rejects a component whose chosen convex-ended boundary edge is shorter than every tile side, which is (iii). The segment-semigroup table is (iii) in full, and the boundary-invariant prune is (iv), with the three tests: integrality of the coefficients, vanishing of $B(\partial R)$ at $z = -dc^{-1}$ modulo $3ab$, and the two character bounds. A missing entry in the finite direction table only disables the last prune at that node. Since each recursive placement removes one tile, the tree is finite; Lemmas 47, 48, 49 and 52 show that exhausting every

branch proves nonexistence. The implementation passes its exact geometry self-tests, including positive controls in both branches with every prune enabled. For $N = 33$ it visits 6479 states with all pruning enabled and 22850 states when the general segment-semigroup table and the boundary-invariant pruning are disabled; the latter run retains the wedge and selected-edge prunes, that is (ii) and (iii) in the weak forms just described.

Definition 55 (Run status). A search run is *exhausted* if all of the following hold: the recursion terminated without reaching a node, time, or memory limit; every enumerated placement was either rejected by one of the necessary conditions of Lemmas 48, 49 and 54 or fully expanded; and no geometric operation reported an unsupported or inconsistent case, that is, every edge length was exactly rational, every placed triangle had exactly the tile area, every subtraction produced simple positively oriented pieces, and the area identity of Lemma 51 held at every node. A run that reaches a limit is *undecided*. A run in which any placement was skipped or any geometric case was reported unsupported is *inconclusive*, whatever verdict it printed, and in particular is never exhausted. Only an exhausted run supports a nonexistence claim. In the implementation each such event increments a warning counter that is part of every archived run record; a record with a nonzero counter is not used in any theorem, and all records cited in this paper report zero. A missing entry in the direction table of the boundary invariant is not such an event and is counted separately.

Repeated components are cached. The equivalence the implementation decides is translation, which is strictly finer than congruence.

Lemma 56 (Canonical states). *Call two components equivalent when one is a translate of the other. Then:*

- (i) *two components are equivalent if and only if they have the same canonical key, where the key of a component is its positively oriented vertex list translated so that its lexicographically least vertex becomes the origin, and then cyclically shifted to the lexicographically least shift;*
- (ii) *tileability is invariant under the equivalence;*
- (iii) *consequently caching tileable and untileable components by their canonical key is sound, and a cached tiling is transported to any translate by the same translation.*

Proof. (i) The construction is translation-equivariant: translating every vertex by t translates the lexicographically least vertex by t and leaves both the difference vectors and the cyclic order unchanged, so translates receive the same key. Conversely, equal keys mean that the two translated vertex lists agree after a cyclic shift, so the two polygons differ by the translation carrying one least vertex to the other.

(ii) A translation is an isometry, and the family of admissible tiles is closed under isometries because congruent copies are allowed to be reflected. So a tiling of R is carried to a tiling of $R + t$ and back.

(iii) Whether a component can be tiled depends on the component alone and not on the tiles already placed elsewhere: by Lemma 52(ii) the pieces produced at a node are independent subproblems. Hence a key that has been shown untileable can be rejected on every later occurrence, and a stored solution can be replayed after translation. $\square$

The key factors out translation and the choice of starting vertex, and nothing else: rotations and reflections are not normalized, although (ii) would justify normalizing them, since the tile set is closed under all isometries. Declining to do so costs cache hits only. The certificate export runs of Appendix A.3 disable the cache entirely, so Theorem 16 and Corollary 13(ii) do not depend on this lemma.

A separately written diagnostic searcher is included for transparency, but its historical output is not part of the proof. A previous version rejected a flush tile edge whenever its length exceeded that of the first boundary segment on the ray. Lemma 47 shows why this restriction is invalid when the next boundary vertex is reflex. The supplied regression reaches a legal two-placement state in which the first segment has length 5 and a contained tile edge has length 7. The cap has been removed; the corrected diagnostic search has not been exhausted, so no independent-exhaustion claim is made.

A.3. Refutation certificates and the independent checker.

The object. A refutation certificate is a finite proof object for the statement that a given target polygon admits no tiling by a given tile. Its format, `erdos634-refutation-v1`, is fixed normatively in `computation/tiling_search/CERT_SPEC.md`; the file is gzipped JSONL, line one a header and every later line a node in depth-first preorder. Arithmetic is exact and symbolic: a coordinate is a pair of rationals (x, w) denoting $x + w\sqrt{D}$, a point a pair of coordinates, a polygon a positively oriented list of at least three points which is simple and has no three consecutive collinear vertices. The header fixes the instance name, the field parameter D , the exact tile side triple, the target polygon in standard position, the pruning configuration, and the number of node lines.

Every node carries its region in *absolute* coordinates, because the export run disables memoization and canonicalization: the object is a tree, no state identification has to be trusted, and Lemma 56 is not used. The export run also uses the *reduced* pruning configuration, whose only rules are the area budget, the wedge test, and the selected-edge test of Lemma 54(i)–(iii); the segment-semigroup table and the boundary-invariant prune are switched off, so a certificate depends on neither Theorem 19 nor the general form of Lemma 53.

A node is either *pruned*, carrying a witness that names one of the three rules and the vertex or edge it applies to but no numerical data — the checker recomputes the claim from the region and the tile alone — or *branched*, carrying a convex vertex, one of the two boundary rays at it, and the complete twelve-element candidate list of Lemma 48(i) in the order fixed there, each entry marked rejected or expanded. An expanded entry carries the child regions in absolute coordinates and singles out one of them as the *refuted child*, with the identifier of the node that refutes it. The remaining children carry no node records: a refutation needs one untileable child, not all of them. An expanded entry with no children is invalid, since by Lemma 45 it would assert that the placed tile alone is the region.

Two design points carry the independence. First, the child regions are re-derived rather than trusted. The checker verifies that every child is contained in the parent, that the children and the placed triangle have pairwise disjoint interiors, that every child edge lies, as a point set, inside the union of the parent’s and the placed triangle’s boundaries — verified by exact one-dimensional coverage along the common supporting line, since a child edge may span a tile edge and a collinear parent edge together — and that the areas of the children plus the tile area equal the parent area. By Lemma 45 the children and the tile then cover the parent exactly, and the edge condition supplies the remaining hypothesis of Lemma 52(ii), so every tile of a hypothetical completion lies in exactly one child. The searcher’s subtraction, that is Lemma 51, is thereby removed from the checked proof. Second, the checker, not the exporter, decides which placements are legal: it constructs the twelve candidates itself from the tile side lengths by the law of cosines and decides containment by the exact test $\text{area}(\Delta \cap R) = \text{area}(\Delta)$, which by Lemma 46 is equivalent to $\Delta \subseteq R$. A candidate that is contained but recorded as rejected, or not contained but recorded as expanded, invalidates the certificate.

The obligations. The checker verifies, in exact arithmetic over $\mathbb{Q}(\sqrt{D})$, and rejects on any failure: (1) that the header is well formed, that the instance is one it knows, that the target

it constructs itself in standard position from the published side lengths equals the recorded target exactly, and that the tile triple satisfies the defining relation of its branch; (2) that the node count matches, that identifiers run $0, \dots, n - 1$ in preorder, and that every node except the root is the refuted child of exactly one earlier node, whose identifier is smaller; (3) that every recorded polygon is simple, positively oriented, of positive area, without repeated or collinear consecutive vertices; (4) that the root region is the target; (5) for each pruned node, the witness claim recomputed from the region and the tile: a non-integral area ratio for the budget rule; for the wedge rule that the interior angle at the named vertex is not a nonempty sum of tile angles, by the bounded enumeration that Lemma 54(ii) justifies; for the selected-edge rule that both endpoints are convex and the length is smaller than every tile side; (6) for each branched node, that the named vertex is convex, that the chosen boundary edge has rational length, and that the twelve candidates it recomputes agree with the recorded ones in order; (7) containment of every candidate, decided by the checker itself, together with the recorded verdict; (8) the child obligations of the previous paragraph, including that the selected child equals the region of the refuted node exactly; (9) that no expanded entry has an empty child list; and (10) acceptance, reported with the instance name, the node count, and the SHA-256 digest of the certificate file. There is no accepted-with-warnings state: any diagnostic is a rejection.

Where the exporter's own tests are weaker than the recomputed claims, the recomputation is the stronger statement and holds a fortiori. The searcher's wedge prune fires when a convex angle is smaller than every tile angle, and such an angle is certainly not a nonempty sum of tile angles. The searcher discards a placement one of whose pieces has non-integral area budget without recursing, by Lemma 54(v); in the certificate that placement is expanded, and the piece in question is recorded as its refuted child and refuted by a node pruned under the budget rule.

Independence. The checker `computation/tiling_search/check_refutation.py` was written from the specification and the lemmas of this appendix alone. It shares no code with `computation/tiling_search/searcher2.py`, with the earlier searchers, or with the positive-certificate verifier `computation/tiling_search/verify_tiling.py`; it uses the Python standard library only, and reimplements the field arithmetic, the polygon predicates, the candidate construction, and the exact polygon intersection areas from scratch. The exporter is a mode of the searcher and may reuse anything in it: it only produces the object, and nothing about how it was produced is trusted.

Theorem 57 (Soundness of certificate checking). *Let a certificate be accepted by a checker discharging obligations (1)–(10), and let T and R_0 be the target polygon and the tile of its header. Then T admits no tiling by congruent copies of R_0 ; reflections are allowed and no edge-to-edge hypothesis is imposed.*

Proof. We prove by induction over the tree, in decreasing order of node identifier, that the region of every node admits no tiling by congruent copies of R_0 . Obligation (2) makes this a valid induction: a node's refuted child has a larger identifier.

Let ν be a node whose region R is, for contradiction, tiled. If ν is pruned then, by obligation (5), the recomputed witness claim holds for R : a non-integral ratio $\text{area}(R)/\text{area}(R_0)$ contradicts Lemma 54(i); a convex interior angle that is not a nonempty sum of tile angles contradicts Lemma 54(ii); and a convex-ended boundary edge shorter than every tile side contradicts Lemma 54(iii).

If ν is branched, let P be its recorded vertex, convex by obligation (6), and let u be the recorded boundary ray at P . By Lemma 47 the tiling contains a tile Δ with a vertex at P

and a complete edge starting at P along u , and by Lemma 48(i) that tile is one of the twelve candidates, which by obligation (6) are exactly the recorded ones. Since $\Delta \subseteq R$, the checker's containment test finds Δ contained (Lemma 46), so by obligation (7) the entry of Δ is marked expanded, and by obligations (8) and (9) it carries children $C_1, \dots, C_k$ with $k \geq 1$ such that: each $C_j \subseteq R$; the sets $\Delta, C_1, \dots, C_k$ have pairwise disjoint interiors; every ∂C_j lies in $\partial R \cup \partial \Delta$; and $\text{area}(\Delta) + \sum_j \text{area}(C_j) = \text{area}(R)$. By Lemma 45, $\Delta \cup \bigcup_j C_j = R$, so by Lemma 52(ii) every tile of the tiling other than Δ lies in exactly one C_j and the tiles lying in C_j tile C_j . In particular the refuted child is tileable. Its region equals the region of the node with the recorded identifier, which is larger than that of ν , so by induction that region admits no tiling: a contradiction.

Hence no node's region is tileable. By obligation (4) the root's region is the header target, so the header target admits no tiling; by obligation (1) that target is the instance's published one, constructed by the checker itself, and its tile triple is the published tile. Obligation (10) ties the accepted object to the archived file by its digest. $\square$

The first campaign.

instance	target and tile	nodes	certificate SHA-256
N33	(21, 21, 33) by (5, 3, 7), $D = 3$	23157	613c52dd...c21919
N21	(12, 12, 21) by (2, 3, 4), $D = 15$	887	70633336...9eaf76

The checker has SHA-256 52d74ab3...7fd99c and accepts both certificates, with total wall-clock runtime under one minute. Certificates are bulk machine output and are not tracked in the repository; their digests, the exact commands, and the interpreter versions are recorded in `paper/verification_manifest.md`, and the full digests are printed there as well. These two instances are the negative computations on which Theorem 16 and Corollary 13(ii) depend. The remaining negative searches in this paper, in particular the twenty-seven exhaustions of Theorem 33 and the searches recorded in Appendix E, are not yet certified; the pipeline applies to them unchanged, and the deferred campaigns and their commands are recorded with the specification in the ancillary files.

A.4. The complete $N = 33$ reduction. The reduction to the searched target is Proposition 14 together with Tables 2 and 3: the sixteen mutually exhaustive branches leave exactly row I for $(a, b, c) = (5, 3, 7)$ at $k = 3$, namely the target with sides (21, 21, 33). This agrees with Beeson's effective isosceles enumeration [3, Theorem 12.10 and Example 2].

Every branch check, including the Group-1 equation sweep, is executable in `computation/verify_theory_report3_claims.py`. The primary exact-search commands, source and output hashes, interpreter version, and stored outputs are recorded in `paper/verification_manifest.md`; the same manifest identifies the diagnostic regression and its scope.

A.5. Verification runs. The following checks were rerun for this manuscript:

check	result
Laurent and signed-character invariant checks	all passed
threshold and auxiliary-ray arithmetic	23/23
branch and fixed- N audit, including the Group-1 equation sweep at 33	90/90
boundary-polynomial and transfer audit	37/37
Group 1 and macro arithmetic audit	18/18
construction and layer arithmetic audit	53/53
addition and conditional-layer arithmetic	27/27
one-patch and lattice bounded regressions	29/29
searcher geometry self-tests	all passed
$N = 33$ fresh primary exhaustion	6479 states, no warnings
$N = 33$ reduced-pruning replay	22850 states, no warnings
$N = 33$ refutation certificate export	23157 nodes
$N = 33$ certificate, independent checker	ACCEPT
$N = 21$ refutation certificate export	887 nodes
$N = 21$ certificate, independent checker	ACCEPT
long-edge/reflex-vertex regression	passed
nine positive coordinate certificates	all passed

What a replay must reproduce. Node and state counts, cut and warning counters, and the SHA-256 digests of certificates and stored outputs are machine-independent invariants of the recorded commands: the searches are deterministic and their arithmetic is exact, so a replay on any platform must reproduce these integers exactly, and a discrepancy is a defect rather than a tolerance. Wall-clock times, node rates, and peak resident memory are environment-dependent and are recorded only as capacity information; the same holds for the checker runtime quoted in Appendix A.3. The pinned environment is recorded in `paper/replay_environment.toml`: the release date, platform, and kernel; the interpreters for the original standard-library search runs (Python 3.9.6, no dependencies) and for the certificate export and checker runs (Python 3.14.6, no dependencies); the interpreter and the pinned `sympy` and `mpmath` versions used for the Laurent audit; the interpreter for the diagnostic regression; and the document build tool. The exact commands and hashes are collected in `paper/verification_manifest.md`. Passing a script verifies only the claims named in that script. In particular, `verify_theory_report6_claims.py` verifies arithmetic consequences of the proposed forced-layer induction, not the missing geometric induction itself. The final audit script performs finite regression checks of the one-patch enumeration, transfer identities, relation lattice, pure-edge word, and defect-wedge formula. Its ranges include $r \leq 35$ for the Group 2 parametrization and defect tests, $p, q < 60$ for Group 1 identities, and support words of length at most 10; the complete bounds are recorded in the script and manifest. The universal statements are proved in the text, not by extrapolation from these tests. The script also checks that the manually transcribed inventory of Herdt’s figure sums to 720, and records an exact mirror-placement witness to the unresolved forced-layer step. It does not certify that figure’s coverage, forced-layer induction, seam propagation, macro-normalization, or semilinearity.

A.6. The trapezoid campaign. For $(a, b, c) = (3, 5, 7)$ and $L = 15$, the seven certificate values are:

x	29	31	32	35	36	38	41
$N = 2x + 15$	73	77	79	85	87	91	97.

Each certificate passes exact congruence, containment, all pairwise intersection, and exact-area checks. The search exhausts the remaining arithmetically possible values in the finite window:

all possible $x \leq 28$, and $x = 30, 33$, together with the other cases not supplied by Lemma 26. The aggregate is 27 exhausted cases and seven tilings, 5,699,235 states, and zero warnings.

The negative half, Theorem 33, is computer-assisted and depends on one exhaustion implementation. Its search algorithm has the completeness argument of Appendix A.2 and extensive local tests, and each of the 27 runs is exhausted in the sense of Definition 55; the seven positive cases, Theorem 32, are independently checkable exact coordinate certificates. The refutation-certificate pipeline of Appendix A.3, which does recompute the complete candidate set and the child regions at every state, applies to these trapezoid targets without modification; exporting and checking the 27 cases is deferred, and the commands are recorded with the format specification. Until that is done, Corollary 34 carries the weaker evidence class of Theorem 33.

APPENDIX B. ROW ALGEBRA AND THE OTHER BRANCHES

B.1. Group 2 target evaluations. Let

$$\varepsilon : \mathbb{Z}[z, z^{-1}] \longrightarrow \mathbb{Z}/(3ab), \quad z \longmapsto -dc^{-1}.$$

Counterclockwise traversal gives the following target polynomials and evaluations:

$$\begin{array}{ll} \text{I : } B = k(U - cz - cz^{-1}), & d\varepsilon(B) \equiv 3a^2k; \\ \text{II : } B = k(3bW - c^2z^{-2} - Ucz), & d^2\varepsilon(B) \equiv 6abkd^2 \equiv 0; \\ \text{III : } B = k(Wc - acz^2 - bVz), & c\varepsilon(B) \equiv 3abdk \equiv 0; \\ \text{IV : } B = k((2a + b) - cz), & \varepsilon(B) \equiv 3ak; \\ \text{V : } B = k(c^2 - (aU + bV)z^2), & c^2\varepsilon(B) \equiv -6abkd^2 \equiv 0; \\ \text{E : } B = \pm 3Lz^{m_0}, & \varepsilon(B) = \pm 3L\varepsilon(z)^{m_0}. \end{array}$$

All congruences are modulo $3ab$. The factors c, d , and $\varepsilon(z)$ used to clear denominators are units there. This proves precisely the conditions in Table 1.

Lemma 58 (Primitivity of the Group 2 target vectors). *Each of the five integer side vectors in rows I–V of Table 1 is primitive.*

Proof. Suppose throughout that a prime ℓ divides every component of the vector in question. For row I, $\ell \mid c, U$ gives $c^2 \equiv 3b^2 \pmod{\ell}$; if $\ell = 3$ this contradicts $3 \nmid c$, and otherwise it forces $\ell \mid a, b$. For row II, $\ell \mid c$; pairwise coprimality and $3 \nmid c$ give $\ell \nmid 3b$, so $\ell \mid W$, whence $c^2 \equiv b^2 \pmod{\ell}$, a contradiction.

For row III the vector is (ac, bV, Wc) . If $\ell \mid a$, then pairwise coprimality gives $\ell \nmid b$, so $\ell \mid bV$ forces $\ell \mid V = 2a + b$ and hence $\ell \mid b$, a contradiction. Thus $\ell \nmid a$, and $\ell \mid ac$ gives $\ell \mid c$. Again $\ell \nmid b$, so $\ell \mid bV$ gives $\ell \mid V$, hence $b \equiv -2a \pmod{\ell}$. The norm equation then yields $0 \equiv c^2 \equiv 3a^2 \pmod{\ell}$; this forces $\ell = 3$, contrary to $3 \nmid c$. Row IV is immediate from $\gcd(a, c) = 1$. Finally, in row V one has $\ell \mid c$ and hence $\ell \nmid a, b$; therefore $\ell \mid U, V$. The identities $2U - V = 3b$ and $2V - U = 3a$ force $\ell = 3$, again contradicting $3 \nmid c$. $\square$

To justify the integral scale used in Table 1, write the classified target side vector as $\lambda \mathbf{v}$, where $\mathbf{v}$ is one of the primitive integer vectors above. Lemma 18 makes every component of $\lambda \mathbf{v}$ integral. Bézout coefficients for the components of $\mathbf{v}$ then express λ as an integer combination of those integral side lengths, so $\lambda \in \mathbb{Z}$.

B.2. Group 1 boundary factorization. Put

$$b = q^2 - p^2, \quad h = 2q^2 - p^2, \quad Q = qz^2 + pz + q, \quad \overline{Q} = qz^2 - pz + q.$$

For the primitive tile (pq, b, q^2) , the two one-tile boundary polynomials factor as

$$F(z) = -pqz^3 + bz^2 + q^2 = (q - pz)Q,$$

$$G(z) = q^2z^3 + bz - pq = (qz - p)Q.$$

The common quadratic is the geometric direction relation. Here z is the character of the direction lattice of this branch: under the standing incommensurable-angle hypothesis $\alpha/\pi \notin \mathbb{Q}$ (rationality of α/π together with $3\alpha + 2\beta = \pi$ would make α/β rational), and every tile and target angle lies in $\mathbb{Z}\pi + \mathbb{Z}\gamma$, namely $\alpha = 2\gamma - \pi$, $\beta = 2\pi - 3\gamma$, and the shape-4 and shape-5 target angles $\alpha + \beta = \pi - \gamma$ and $\alpha + 2\beta = 3\pi - 4\gamma$. Every directed edge direction therefore has a unique expression $\theta = j\pi + m\gamma$, $j \in \mathbb{Z}/2$, $m \in \mathbb{Z}$, and with $\chi(\theta) = (-1)^j z^m$ the subdivision-and-cancellation argument in the proof of Theorem 19 applies verbatim: every tile contributes $\pm z^k F$ or $\pm z^k G$, so, with $J = (q - pz, qz - p)$,

$$B(\partial T) \in (F, G) = Q \cdot J \subseteq \mathbb{Z}[z, z^{-1}],$$

the ideal membership used for both shapes below. The homomorphism

$$\mathbb{Z}[z, z^{-1}] \longrightarrow \mathbb{Z}/b, \quad z \longmapsto qp^{-1},$$

is well defined because p and q are units modulo b , and it kills both generators of J . Conversely,

$$q(q - pz) + p(qz - p) = b$$

and $q - pz = 0$ determines every Laurent monomial after reduction modulo b . Hence

$$\mathbb{Z}[z, z^{-1}]/J \cong \mathbb{Z}/b, \quad J \cap \mathbb{Z} = (b).$$

For shape 4, boundary integrality and $\gcd(p, q) = 1$ make the target sides (Cq, Cq, Cp) with $C \in \mathbb{Z}$. Direct insertion gives

$$zB(\partial T) = CQ.$$

Every tile contribution lies in QJ . Cancelling Q in the Laurent domain gives $C \in J$, and hence $C \in J \cap \mathbb{Z} = (b)$. Write $C = bm$. The equal target side is therefore

$$X = Cq = bqm.$$

Attaching a tile scaled by qm along that side produces the shape-1 target used in Proposition 38; Corollary 37 then forces $q \mid m$.

For shape 5, boundary integrality and $\gcd(q^2, h) = 1$ make the target sides (Cq^2, Cq^2, Ch) with $C \in \mathbb{Z}$. Direct insertion gives

$$z^2 B(\partial T) = C(q^2 z^4 + hz^2 + q^2) = CQ\overline{Q}.$$

Every tile contribution lies in QJ . Cancelling Q in the Laurent domain therefore gives $C\overline{Q} \in J$. In the quotient,

$$\overline{Q}(qp^{-1}) = q^3 p^{-2},$$

which is a unit, so $b \mid C$. Writing $C = bn$ in the area identity $N = C^2 h/b$ gives $N = bhn^2$, with no additional parity factor even when p, q are both odd.

B.3. The $\gamma = 2\alpha$ insertion. Put $b = v^2 - u^2$ and $Q(z) = uz^2 - vz + u$. First, $\alpha/\pi \notin \mathbb{Q}$: otherwise $2\cos\alpha = v/u$ would be a rational algebraic integer and hence an integer, whereas $1 < v/u < 2$ because $u < v < 2u$.

Since $\beta = \pi - 3\alpha$ and $\gamma = 2\alpha$, every angle of the tile and of the target lies in $\mathbb{Z}\pi + \mathbb{Z}\alpha$, so in a tiling of this branch every directed edge direction has, once an anchor is chosen, a unique expression

$$\theta = j\pi + m\alpha, \quad j \in \mathbb{Z}/2, \quad m \in \mathbb{Z};$$

uniqueness follows from $\alpha/\pi \notin \mathbb{Q}$ exactly as in Section 4. Define the character $\chi(\theta) = (-1)^j z^m \in \mathbb{Z}[z, z^{-1}]^\times$ and the boundary value $B(PQ) = |PQ| \chi(\theta)$ as there. Anchor the c -edge of a direct tile at $\theta = 0$: traversing the edges c, a, b counterclockwise, the exterior angles $\pi - \beta = 3\alpha$ and $\pi - \gamma$ place them in directions $0, 3\alpha, \pi + \alpha$, so the two one-tile boundary polynomials are, up to Laurent units,

$$\begin{aligned} F(z) &= u^2 z^3 - bz + uv = (uz + v)Q(z) && \text{(direct),} \\ G(z) &= uvz^3 - bz^2 + u^2 = (u + vz)Q(z) && \text{(reflected);} \end{aligned}$$

the reflected value arises from F by $z \mapsto z^{-1}$ and orientation reversal, and equals G up to the Laurent unit $-z^{-3}$. The subdivision-and-cancellation argument in the proof of Theorem 19 applies verbatim: every tile contributes $\pm z^k F$ or $\pm z^k G$, and interior segments cancel in pairs, so

$$(6) \quad B(\partial T) \in (F, G) = Q \cdot J, \quad J = (uz + v, u + vz),$$

an ideal membership in $\mathbb{Z}[z, z^{-1}]$. For the residual ideal J ,

$$v(uz + v) - u(u + vz) = b, \quad \mathbb{Z}[z, z^{-1}]/J \cong \mathbb{Z}/b, \quad z \mapsto -vu^{-1},$$

the evaluation being well defined because $\gcd(u, b) = 1$; it kills both generators, so $J \cap \mathbb{Z} = (b)$.

The target is isosceles with equal side X and base Y , and the law of sines gives $X = ut$, $Y = vt$ for a real scale $t > 0$. Anchored with its base at $\theta = 0$, its counterclockwise boundary value is

$$B(\partial T) = vt - utz - utz^{-1} = -tz^{-1}Q(z).$$

By (6) there is an $h \in J$ with $-tz^{-1}Q = Qh$; the Laurent ring over $\mathbb{R}$ is a domain, so $-tz^{-1} = h$. In particular the coefficient t of the monomial $-tz^{-1} = h \in \mathbb{Z}[z, z^{-1}]$ is an integer—as it also is by Lemma 18 and Bézout, since $X = ut$ and $Y = vt$ are integers and $\gcd(u, v) = 1$ —and $t = -zh \in J$. Hence

$$t \in J \cap \mathbb{Z} = (b), \quad t = br, \quad r \in \mathbb{Z}, \quad r \geq 1.$$

This is the force of the ideal membership: it yields $b \mid t$, not merely $t \in \mathbb{Q}$. Beeson's area equation $X^2 = Nu^2b$ (Theorem 9) then gives $N = u^2t^2/(u^2b) = br^2$ with r an integer, the ray of Proposition 40.

Lemma 59 (Angle representations for $\gamma = 2\alpha$). *Let $\gamma = 2\alpha$ and $\beta = \pi - 3\alpha$ with $\alpha/\pi \notin \mathbb{Q}$.*

- (i) *The representations of α , $\alpha + \beta$, and π in the form $x\alpha + y\beta + z\gamma$ with $x, y, z \in \mathbb{N}_0$ are exactly*

angle	(x, y, z)
α	$(1, 0, 0)$
$\alpha + \beta$	$(1, 1, 0)$
π	$(3, 1, 0), (1, 1, 1)$

The first two are unique; π has exactly the two listed representations. In particular $y = 1$ in every representation of $\alpha + \beta$ and of π , and $z \leq 1$ throughout.

- (ii) *Consequently, in a tiling of the isosceles target of Proposition 40, whose base angles are α and whose apex angle is $\alpha + \beta$: a vertex interior to a straight boundary side carries exactly one β sector and at most one γ sector; a base corner carries exactly one sector, an α ; and the apex carries exactly two sectors, one α and one β , so exactly two tiles meet there and no target corner carries a γ .*

Proof. Since $\beta = \pi - 3\alpha$ and $\gamma = 2\alpha$,

$$x\alpha + y\beta + z\gamma = (x - 3y + 2z)\alpha + y\pi,$$

and α, π are linearly independent over $\mathbb{Q}$ because $\alpha/\pi \notin \mathbb{Q}$; so the two coefficients are determined by the angle represented. For α this gives $y = 0$ and $x + 2z = 1$, whence $(x, z) = (1, 0)$. For $\alpha + \beta = \pi - 2\alpha$ it gives $y = 1$ and $x - 3 + 2z = -2$, again $x + 2z = 1$ and $(x, z) = (1, 0)$. For π it gives $y = 1$ and $x - 3 + 2z = 0$, that is $x + 2z = 3$, whence $(x, z) \in \{(3, 0), (1, 1)\}$; these are the identities $3\alpha + \beta = \pi$ and $\alpha + \beta + \gamma = \pi$. This proves (i).

For (ii), first identify the target angles. The target is isosceles with $X/Y = u/v$, so if θ denotes its base angle, then

$$\frac{u}{v} = \frac{X}{Y} = \frac{\sin \theta}{\sin(\pi - 2\theta)} = \frac{1}{2 \cos \theta},$$

whence $2 \cos \theta = v/u = 2 \cos \alpha$ and $\theta = \alpha$, the cosine being injective on $(0, \pi)$. The apex angle is therefore $\pi - 2\alpha = \alpha + \beta$. The sectors of the tiles incident with a vertex interior to a straight boundary side have angles summing to π , and those at a target corner sum to that corner angle; each sector is a single tile angle. Reading off the table in (i) gives the three assertions, the count of tiles at the apex included. $\square$

It remains to exclude $r = 1$. Then an equal target side has length ub and, by Lemma 18, decomposes into complete tile edges as

$$ub = Au^2 + Bb + Cuv, \quad A, B, C \in \mathbb{N}_0.$$

Reduction modulo u gives $u \mid B$, since $\gcd(u, b) = 1$, while the size bound gives $B \leq u$. Hence $B = 0$ or $B = u$.

Consider either equal side, and write its decomposition as $E = A + B + C$ edges with endpoints $v_0, \dots, v_E$ in order, so that v_0 is a base corner, v_E is the apex, and $v_1, \dots, v_{E-1}$ are interior to the side. By Lemma 59(ii), each v_i with $0 < i < E$ carries exactly one β sector and at most one γ sector. Consequently, for either of the labels β and γ , two edges of the decomposition cannot both place a sector with that label at a common endpoint: adjacent edges would put two β , respectively two γ , sectors at one straight vertex, and nonadjacent edges have no common endpoint. Hence, for either label, sending an edge to its endpoint carrying that label is injective.

If $B = u$, equality of lengths forces $A = C = 0$, so the side consists of $E = u$ edges of length b . The side b is opposite β , so the two endpoints of a b -edge carry α and γ : each of the u edges has exactly one γ endpoint. No target corner carries a γ , so these u endpoints lie among the $u - 1$ interior vertices, contradicting injectivity.

If $B = 0$, then $E = A + C \geq 1$. The sides u^2 and uv are opposite α and γ , so the endpoints of an a -edge carry β, γ and those of a c -edge carry α, β : each of the E edges has exactly one β endpoint. Exactly two tiles meet at the apex, one contributing an α sector and one a β sector, and each of the two equal sides is adjacent to one of them; choose the equal side whose apex tile contributes the α . Then neither of its two corners receives a β endpoint: the base corner carries no β sector at all, and the edge of the decomposition emanating from the apex is a complete edge of the apex tile, whose sector there is α , so that edge carries its β at its other

endpoint. Hence the E beta endpoints lie among the $E - 1$ interior vertices, again contradicting injectivity. Thus $r \geq 2$. The argument does not imply $r \geq 3$.

APPENDIX C. ADDITION DATA AND CONSTRUCTIVE TRANSFERS

For Proposition 27, the following corner data work at multiplier one:

ray	target adjacent sides	tile adjacent sides	integer ratios
I	bc, bU	c, b	(b, U)
II	$Uc, 3bW$	c, b	$(U, 3bW)$
III	bV, cW	b, c	(V, W)
IV	bc, bW	c, b	(b, W)
V	aU, bV	a, b	(U, V) , supplement
E	ab, ab	a, b	(b, a) , supplement
G1 shape 4	bc, bc	b, c	(c, b)
G1 shape 5	$bc, b(2q^2 - p^2)$	c, b	$(b, 2q^2 - p^2)$

For the row-V interval transfer, bisect the two doubled base angles of the row-V target at multiplier s . Label its vertices

$$\angle A = 2\beta, \quad \angle B = 2\alpha, \quad \angle C = \alpha + \beta = \pi/3,$$

and let the bisectors from A and B meet at I . The sine rule in $\triangle AIB$, which has angles $\beta, \alpha, 2\pi/3$, gives $AI = acs$ and $BI = bcs$. Thus AIB is a cs -scaled copy of the tile.

Choose $P \in AC$ so that AIP is an as -scaled tile with AI as its c -side, and choose $Q \in BC$ analogously so that BIQ is a bs -scaled tile. The angles at I are respectively $\alpha, 2\pi/3, \beta$, so P, I, Q are collinear. Moreover $PI = IQ = abs$, hence $PQ = 2abs$. The angles at P and Q supplementary to the two $2\pi/3$ tile angles are both $\pi/3$, as is $\angle C$; therefore PCQ is equilateral of side $2abs$.

Quadratically tiling the three corner triangles and tiling the equilateral core gives

$$(c^2 + a^2 + b^2 + 4ab)s^2 = UVs^2,$$

where the equality uses $c^2 = a^2 + ab + b^2$. Proposition 30 then gives the same multiplier interval for row II.

C.1. The complete one-patch tables. We use ordered notation

$$\langle \theta_0, \theta_1, \theta_2 \mid x_0, x_1, x_2 \rangle,$$

where x_i is opposite θ_i . The elementary gluing rule used below is

$$\begin{aligned} & \langle \theta_0, \theta_1, \theta_2 \mid x_0, x_1, x_2 \rangle \cup \langle \phi_0, \phi_1, \phi_2 \mid ty_0, ty_1, ty_2 \rangle \\ &= \langle \theta_0, \theta_2 + \phi_2, \phi_0 \mid ty_1, x_2 + ty_2, x_1 \rangle, \end{aligned}$$

provided $x_0 = ty_0$, the corresponding seam endpoints satisfy $\theta_1 + \phi_1 = \pi$, and $\theta_2 + \phi_2 < \pi$. Thus one endpoint disappears, while the two collinear outer sides at that endpoint merge by addition.

The ordered Group 2 data are

row	angles	opposite sides
$I(m)$	$(\alpha, \alpha, \alpha + 3\beta)$	$m(bc, bc, bU)$
$II(m)$	$(\alpha, 2\alpha, 3\beta)$	$m(c^2, Uc, 3bW)$
$III(m)$	$(\alpha, 2\beta, 2\alpha + \beta)$	$m(ac, bV, Wc)$
$IV(m)$	$(\alpha, \alpha + \beta, \alpha + 2\beta)$	$m(ab, bc, bW)$
$V(m)$	$(2\alpha, 2\beta, \alpha + \beta)$	$m(aU, bV, c^2)$
$E(m)$	$(\alpha + \beta, \alpha + \beta, \alpha + \beta)$	$m(ab, ab, ab)$

and, with $h = b + c$ and $g = b + 2c$, the ordered Group 1 data are

shape	angles	opposite sides
$S_1(n)$	$(2\alpha, \beta, \alpha + \beta)$	$n(ah, bc, c^2)$
$S_2(n)$	$(2\alpha, \alpha, 2\beta)$	$n(ch, c^2, bg)$
$S_3(n)$	$(\beta, \beta, 3\alpha)$	$n(c^2, c^2, ag)$
$S_4(n)$	$(\alpha + \beta, \alpha + \beta, \alpha)$	$n(bc, bc, ab)$
$S_5(n)$	$(\alpha, \alpha, \alpha + 2\beta)$	$n(bc, bc, bh)$

For Group 2, the attachment scales and count identities in Proposition 42 are:

transfer	patch scale	normalized count identity
$E(m) \rightarrow IV(m)$	bm	$ab + b^2 = bW$
$IV(m) \rightarrow I(m)$	bm	$bW + b^2 = bU$
$IV(m) \rightarrow III(b, a; m)$	Wm	$bW + W^2 = UW$
$V(m) \rightarrow II(m)$	Um	$UV + U^2 = 3UW$
$IV(as) \rightarrow IV(cs)$	bWs	$bWa^2 + b^2W^2 = bWc^2$
$V(as) \rightarrow III(a, b; cs)$	bVs	$UVa^2 + b^2V^2 = VWc^2$
$V(bs) \rightarrow III(b, a; cs)$	aUs	$UVb^2 + a^2U^2 = UWc^2$

Here and below the omitted parameter pair is (a, b) . The last two identities follow from

$$Ua^2 + b^2V = Wc^2$$

and its $a \leftrightarrow b$ image.

In every case the straight pair is $(\alpha + \beta) + \gamma = \pi$. The shared-side and merged-side checks are:

transfer	shared side	other angle	destination sides
$E(m) \rightarrow IV(m)$	$abm = a(bm)$	$(\alpha + \beta) + \beta$	$(abm, bcm, \underbrace{abm + b^2m}_{bWm})$
$IV(m) \rightarrow I(m)$	$abm = a(bm)$	$(\alpha + 2\beta) + \beta$	$(bcm, bcm, \underbrace{bWm + b^2m}_{bUm})$
$IV(m) \rightarrow III(b, a; m)$	$bWm = b(Wm)$	$\alpha + \alpha$	$(bcm, \underbrace{abm + aWm}_{aUm}, cWm)$
$V(m) \rightarrow II(m)$	$aUm = a(Um)$	$2\beta + \beta$	$(c^2m, cUm, \underbrace{bVm + bUm}_{3bWm})$
$IV(as) \rightarrow IV(cs)$	$abWs = a(bWs)$	$\alpha + \beta$	$(abcs, \underbrace{a^2bs + b^2Ws}_{bc^2s}, bcWs)$
$V(as) \rightarrow III(a, b; cs)$	$abVs = a(bVs)$	$2\alpha + \beta$	$(ac^2s, bcVs, \underbrace{a^2Us + b^2Vs}_{Wc^2s})$
$V(bs) \rightarrow III(b, a; cs)$	$abUs = b(aUs)$	$2\beta + \alpha$	$(bc^2s, acUs, \underbrace{b^2Vs + a^2Us}_{Wc^2s})$

For Group 1 put

$$a = pq, \quad b = q^2 - p^2, \quad c = q^2, \quad h = b + c, \quad g = b + 2c.$$

The normalized tables are:

transfer	patch scale	normalized count identity
$S_4(n) \rightarrow S_1(n)$	cn	$bc + c^2 = ch$
$S_4(n) \rightarrow S_5(n)$	bn	$bc + b^2 = bh$
$S_1(n) \rightarrow S_2(n)$	hn	$ch + h^2 = hg$
$S_1(n) \rightarrow S_3(n)$	cn	$ch + c^2 = cg$
$S_1(bs) \rightarrow S_1(cs)$	ahs	$chb^2 + a^2h^2 = chc^2$
$S_4(ps) \rightarrow S_4(qs)$	bqs	$bcp^2 + b^2q^2 = bcq^2$

The corresponding Group 1 side checks are:

transfer	shared side	other angle	destination sides
$S_4(n) \rightarrow S_1(n)$	$bcn = b(cn)$	$\alpha + \alpha$	$(\underbrace{abn + acn}_{ahn}, bcn, c^2n)$
$S_4(n) \rightarrow S_5(n)$	$abn = a(bn)$	$(\alpha + \beta) + \beta$	$(bcn, bcn, \underbrace{bcn + b^2n}_{bhn})$
$S_1(n) \rightarrow S_2(n)$	$ahn = a(hn)$	$\beta + \beta$	$(chn, c^2n, \underbrace{bcn + bhn}_{bgn})$

transfer	shared side	other angle	destination sides
$S_1(n) \rightarrow S_3(n)$	$bcn = b(cn)$	$2\alpha + \alpha$	$(c^2n, c^2n, \underbrace{ahn + acn}_{agn})$
$S_1(bs) \rightarrow S_1(cs)$	$abhs = b(ahs)$	$\beta + \alpha$	$(achs, bc^2s, \underbrace{b^2cs + a^2hs}_{c^3s})$
$S_4(ps) \rightarrow S_4(qs)$	$bcp s = a(bqs)$	$\alpha + \beta$	$(bcqs, \underbrace{abps + b^2qs}_{bcqs}, \underbrace{bcps}_{abqs})$

The final identities use

$$b^2c + a^2h = c^3, \quad abp + b^2q = bcq, \quad bcp = abq.$$

For completeness, the exhaustion is an angle calculation, not a search over lengths. In Group 2,

$$\pi = 3\alpha + 3\beta, \quad \gamma = 2\alpha + 2\beta.$$

The complements in π of the three tile angles are respectively $2\alpha + 3\beta$, $3\alpha + 2\beta$, and $\alpha + \beta$. Among target vertices capable of a straightening attachment, only $\alpha + \beta$ occurs, in rows E, IV, and V. Assigning the two remaining tile angles to the two seam endpoints gives exactly the Group 2 rows above, their reflected forms, and one IV-to-reptile output. Similarly, in Group 1 the complements are $2\alpha + 2\beta$, $3\alpha + \beta$, and $\alpha + \beta$. Only $\alpha + \beta$ occurs, in S_1 and S_4 ; the two assignments give the six listed transfers and the S_1 -to-reptile and S_4 -to-reptile outputs. The incommensurability hypothesis ensures that no unlisted equality of angle vectors is hidden in these comparisons.

More explicitly, let $\mathcal{O}_j(X)$ be the set of row-shaped angle outputs from a source row X in Group j , let R denote a tile-shaped output, and let $*$ interchange (a, α) with (b, β) . The enumeration is

$$\begin{aligned} \mathcal{O}_2(I) &= \mathcal{O}_2(II) = \mathcal{O}_2(III) = \emptyset, \\ \mathcal{O}_2(E) &= \{IV, IV^*\}, \quad \mathcal{O}_2(IV) = \{I, IV, III^*, R\}, \\ \mathcal{O}_2(V) &= \{II, II^*, III, III^*\}, \end{aligned}$$

and

$$\begin{aligned} \mathcal{O}_1(S_2) &= \mathcal{O}_1(S_3) = \mathcal{O}_1(S_5) = \emptyset, \\ \mathcal{O}_1(S_1) &= \{S_1, S_2, S_3, R\}, \quad \mathcal{O}_1(S_4) = \{S_1, S_4, S_5, R\}. \end{aligned}$$

For the five non-preserving angle outputs, the generic destination multipliers are

$$\frac{cm}{a}, \quad \frac{cm}{a}, \quad \frac{cm}{b}, \quad \frac{cm}{b}, \quad \frac{qm}{p}.$$

Pairwise coprimality of (a, b, c) and $\gcd(p, q) = 1$ force, respectively, $a \mid m, a \mid m, b \mid m, b \mid m, p \mid m$. These are exactly the as, as, bs, bs, ps parametrizations in Proposition 42; the remaining outputs preserve the normalized multiplier.

C.2. Orientation-changing parallelograms and the 720 inventory.

Lemma 60 (Herdt–Beeson parallelogram). *Let $p, q \in \mathbb{N}_0$, not both zero. A parallelogram with adjacent side lengths*

$$bc, \quad pc + qb$$

and included angle α is tiled by $2(pc + qb)$ copies of R . If $p, q > 0$, its two subparallelograms use different grid orientations. Cyclic side-label variants hold as well.

Proof. Split the second side after length pc . The first subparallelogram is a c -by- p array of fundamental cells with adjacent sides b, c ; the second is a b -by- q array with those roles reversed. Each cell splits along its a -diagonal into two copies of R . The shared line may have different edge subdivisions on its two sides, which is permitted because tilings need not be edge-to-edge. $\square$

For $(a, b, c) = (4, 5, 6)$, Herdt's 720-tiling in [3, Fig. 24] has target sides $(120, 120, 180)$ and count $5 \cdot 12^2$. Its vector drawing has ten triangular macroregions: one quadratic patch of scale 15 and three each of scales 6, 5, 4. These contribute

$$15^2 + 3(6^2 + 5^2 + 4^2) = 456.$$

The inventory has three orientation-changing blocks. Two have common side 30 and width $59 = 24 + 35$, with uniform constituent counts $48 + 70$ in opposite orders; the third is the cyclically relabelled block with common side 20 and width $14 = 4 + 10$, with counts $8 + 20$. Thus the macro count is

$$456 + (48 + 70) + (48 + 70) + (8 + 20) = 720.$$

The rounded source vector paths are consistent with this grouping and total area. Since no exact labelled macro coordinates are archived, this paragraph records an arithmetic audit of a manually transcribed figure inventory rather than a coordinate or coverage certificate.

C.3. The shape-1/shape-5 complement. Every shape-4 tiling has $m = qn$ and, after a single scaled tile is attached along its base, produces a shape-5 tiling at multiplier n . Thus shape 4 is subordinate to shape 5. Conversely, a shape-1 and a shape-5 target at the same multiplier glue along a common side to form a reptiling; the remaining difficulty is to force the corresponding seam inside an arbitrary reptiling.

APPENDIX D. ORDERED INTERIOR STRUCTURES

The relation lattice. Let $\rho = e^{i\pi/3}$, so

$$N(x + y\rho) = x^2 + xy + y^2.$$

Lemma 61 (Primitive Eisenstein-square lemma). *Let (a, b, c) be a primitive positive solution of $c^2 = a^2 + ab + b^2$. Then, after possibly interchanging a and b , there are integers $r > s > 0$ with $\gcd(r, s) = 1$ and $r \not\equiv s \pmod{3}$ such that*

$$a = r^2 - s^2, \quad b = 2rs + s^2, \quad c = r^2 + rs + s^2.$$

Conversely, every pair of integers $r > s > 0$ with $\gcd(r, s) = 1$ and $r \not\equiv s \pmod{3}$ yields by these formulae a primitive positive solution.

Proof. Work in the ring of Eisenstein integers $\mathbb{Z}[\rho]$, a Euclidean domain and hence a unique factorization domain. Here $\rho^2 = \rho - 1$, $\bar{\rho} = 1 - \rho = \rho^{-1}$, the norm $N(x + y\rho) = x^2 + xy + y^2$ is multiplicative and positive definite, and the unit group is the cyclic group $\mathbb{Z}[\rho]^\times = \langle \rho \rangle$ of order six, with $\rho^3 = -1$. Put

$$\xi_0 = a + b\rho, \quad \bar{\xi}_0 = a + b - b\rho, \quad \xi_0 \bar{\xi}_0 = N(\xi_0) = c^2.$$

A common prime divisor lies above 3. Let π be a prime of $\mathbb{Z}[\rho]$ dividing both ξ_0 and $\bar{\xi}_0$. From $\rho - \bar{\rho} = 2\rho - 1$ and $\rho + \bar{\rho} = 1$,

$$\xi_0 - \bar{\xi}_0 = b(2\rho - 1), \quad \xi_0 + \bar{\xi}_0 = 2a + b,$$

so $\pi \mid b(2\rho - 1)$ and $\pi \mid 2a + b$. Now $N(2\rho - 1) = (-1)^2 + (-1) \cdot 2 + 2^2 = 3$, so $2\rho - 1$ is prime; it is the ramified prime above 3, since $(2\rho - 1)^2 = 4\rho^2 - 4\rho + 1 = -3$. If π is not associate to

$2\rho - 1$, then $\pi \mid b$, hence $\pi \mid a$ because $\pi \mid a + b\rho$; writing $1 = ua + vb$ with $u, v \in \mathbb{Z}$, available since $\gcd(a, b) = 1$ by Lemma 10, gives $\pi \mid 1$, which is absurd. So π is associate to $2\rho - 1$.

Primitivity and $3 \nmid c$ exclude that prime. If $2\rho - 1$ divided ξ_0 , then $3 = N(2\rho - 1)$ would divide $N(\xi_0) = c^2$ and hence $3 \mid c$, contradicting Lemma 10. Therefore ξ_0 and $\bar{\xi}_0$ have no common prime divisor.

All prime exponents are even. For every prime π of $\mathbb{Z}[\rho]$,

$$v_\pi(\xi_0) + v_\pi(\bar{\xi}_0) = v_\pi(c^2) = 2v_\pi(c),$$

and by the previous step at most one of the two summands is nonzero; hence $v_\pi(\xi_0)$ is 0 or $2v_\pi(c)$, in either case even. Unique factorization therefore gives $\xi_0 = u\xi^2$ with $u \in \mathbb{Z}[\rho]^\times$ and $\xi \in \mathbb{Z}[\rho]$.

Normalizing the unit. The squares of units form the index-two subgroup $\{1, \rho^2, \rho^4\}$ of $\langle \rho \rangle$; note $-1 = \rho^3$ is not a square. Replacing ξ by $\rho^k \xi$ we may therefore assume $u \in \{1, \rho\}$. Since $\rho\bar{\rho} = N(\rho) = 1$ and

$$\overline{\rho a + b\rho} = \rho(a + b - b\rho) = b + a\rho,$$

multiplying the conjugate by ρ is exactly the interchange of a and b . Hence if $u = \rho$, then

$$b + a\rho = \rho\bar{\xi}_0 = \rho\overline{\rho\xi^2} = \rho\bar{\rho}\bar{\xi}^2 = \bar{\xi}^2,$$

so after interchanging a and b we may assume $u = 1$, that is $a + b\rho = \xi^2$.

The chamber. Write $\xi = r + s\rho$ with $r, s \in \mathbb{Z}$. Using $\rho^2 = \rho - 1$,

$$\xi^2 = (r^2 - s^2) + (2rs + s^2)\rho,$$

so $a = r^2 - s^2$ and $b = 2rs + s^2$; moreover $N(\xi) = r^2 + rs + s^2$ is positive with $N(\xi)^2 = N(\xi_0) = c^2$ and $c > 0$, so $c = r^2 + rs + s^2$. Replacing ξ by $-\xi$ leaves ξ^2 unchanged, so we may assume $s \geq 0$. Then $b = s(2r + s) > 0$ forces $s > 0$ and $2r + s > 0$, while $a = r^2 - s^2 > 0$ forces $|r| > s$. If $r < -s$, then $r < -s < -s/2$, contradicting $2r + s > 0$; hence $r > s > 0$. A common prime divisor of r and s would divide both a and b , so $\gcd(r, s) = 1$. Finally, if $r \equiv s \pmod{3}$, then $3 \mid r - s$ gives $3 \mid a = (r - s)(r + s)$, while $2r + s \equiv 2s + s = 3s \equiv 0 \pmod{3}$ gives $3 \mid b = s(2r + s)$, again contradicting $\gcd(a, b) = 1$. Hence $r \not\equiv s \pmod{3}$.

Converse. Let $r > s > 0$ be coprime with $r \not\equiv s \pmod{3}$ and define a, b, c by the displayed formulae. Then $a, b, c > 0$, and $a + b\rho = (r + s\rho)^2$ with $N(r + s\rho) = c$ gives $c^2 = a^2 + ab + b^2$. Let a prime ℓ divide a and b . If $\ell \mid r$, then $\ell \mid a = r^2 - s^2$ forces $\ell \mid s$; if $\ell \mid s$, then $\ell \mid a$ forces $\ell \mid r$; both contradict $\gcd(r, s) = 1$. So $\ell \nmid rs$, and $\ell \mid b = s(2r + s)$ gives $\ell \mid 2r + s$. Also $\ell \mid a = (r - s)(r + s)$. If $\ell \mid r + s$, then $\ell \mid (2r + s) - (r + s) = r$, excluded. If $\ell \mid r - s$, then $\ell \mid (2r + s) - 2(r - s) = 3s$, so $\ell = 3$ and $3 \mid r - s$, contrary to hypothesis. Hence $\gcd(a, b) = 1$, and a fortiori $\gcd(a, b, c) = 1$. $\square$

Lemma 61 is the parametrization asserted in Proposition 43.

Direct expansion now gives

$$rc = (r + s)a + sb, \quad rb = s(a + c), \quad (r + 2s)a = (r - s)(b + c).$$

Thus

$$R_c = (r + s, s, -r), \quad R_b = (s, -r, s)$$

lie in the integer kernel of (a, b, c) . Moreover,

$$R_c \times R_b = -(a, b, c).$$

The normal vector on the right is primitive. Hence the parallelogram spanned by R_c, R_b has index one in the rank-two lattice $\ker_{\mathbb{Z}}(a, b, c)$, proving Proposition 43(i). This statement

concerns signed, unordered counts of complete a -, b -, and c -edges. It contains no endpoint order, connectivity, or geometric-realizability assertion.

Pure-edge support words. For an oriented a -edge $[x, x + a]$ supporting a tile in the upper half-plane, the two configurations in Proposition 43 have endpoint labels

$$A : (\beta, \gamma), \quad B : (\gamma, \beta).$$

An $A \rightarrow B$ junction would put two $\gamma = 2\pi/3$ sectors in the same half-plane and is impossible. Therefore a support word contains no substring AB , which is equivalent to the form $B^j A^{n-j}$.

At an $A \rightarrow A$ or $B \rightarrow B$ junction, the supported angles sum to $\beta + \gamma = \pi - \alpha$, leaving an α sector. If that sector contains $n_\alpha, n_\beta, n_\gamma$ tile angles, then

$$(n_\alpha - n_\beta - 1)\alpha + \frac{n_\beta + 2n_\gamma}{3}\pi = 0,$$

because $\beta = \pi/3 - \alpha$ and $\gamma = 2\pi/3$. The irrationality of α/π , proved in the next paragraph, gives $n_\alpha = 1$ and $n_\beta = n_\gamma = 0$. This forces one tile angle, but it does not decide which of that tile's two adjacent sides lies on which wall; that distinction is the forced-layer mirror gap.

At the possible $B \rightarrow A$ junction, put the common endpoint at $O = 0$. The two supported third vertices are

$$Q_- = -a + b\rho^2, \quad Q_+ = a + b\rho.$$

Using $\rho^2 = \rho - 1$ gives

$$|OQ_-| = |OQ_+| = c, \quad |Q_-Q_+| = 2a + b.$$

Both third vertices have height $b\sqrt{3}/2$, so the defect area is $(2a + b)b\sqrt{3}/4$, whereas the tile area is $ab\sqrt{3}/4$. Their ratio is $(2a + b)/a$. If this were integral, then $a \mid b$; primitivity would force $a = 1$. But then

$$(2c - (2b + 1))(2c + (2b + 1)) = 3$$

forces $b = 0$, contrary to positivity. The defect wedge is therefore never a union of whole tiles. The directed c -flow. Orient every c -edge from its α endpoint to its β endpoint, splitting a through-edge at each T-junction. At a non-corner vertex, let A, B, G count incident α, β, γ tile corners, and let F count incident tiles for which the vertex lies in the relative interior of a tile edge. Each such tile contributes a flat sector of angle π . Since $\gamma = 2(\alpha + \beta)$ and $\pi = 3(\alpha + \beta)$, the angle equation is

$$(A + 2G)\alpha + (B + 2G)\beta = (6 - 3F)(\alpha + \beta) \quad \text{or} \quad (3 - 3F)(\alpha + \beta),$$

according as the vertex is interior or lies straight on the target boundary. Here α/π is irrational. Indeed, $2 \cos \alpha = U/c$ is rational and lies strictly between 1 and 2, because $U^2 - c^2 = 3bW > 0$ and $4c^2 - U^2 = 3a^2 > 0$. If α/π were rational, $2 \cos \alpha$ would be a rational algebraic integer and hence an integer, a contradiction. Hence α and $\beta = \pi/3 - \alpha$ are linearly independent over $\mathbb{Q}$, and $A = B$. An α corner contributes one outgoing c -end, a β corner one incoming end, and a γ corner neither. A through- c edge contributes one incoming and one outgoing end, while a through- a or through- b edge contributes neither. The flow is therefore conserved. At a convex target corner no flat sector can occur. If its angle is $P\alpha + Q\beta$, then

$$A + 2G = P, \quad B + 2G = Q,$$

so its net outflow is $A - B = P - Q$. Standard decomposition of the resulting finite directed multigraph gives directed cycles and source-to-sink c -edge paths. The latter may run along the target boundary; no interiority claim is made.

The row-III half-parallelogram. Let $\rho = e^{i\pi/3}$, $u = b$, and $v = b + a\rho$; then $|v| = c$ and the (u, v) unit parallelogram is two copies of the tile because $v - u = a\rho$. The multiplier-one row-III target is

$$T = \text{conv}(0, Vu, Wv).$$

Indeed, its first two side lengths are bV and cW , while

$$Wv - Vu = a(W\rho - b), \quad |W\rho - b|^2 = W^2 - Wb + b^2 = c^2,$$

so its third side is ac . Its area is VW tile areas. Thus T is one half of the V -by- W parallelogram tiled in the (u, v) grid. Moreover the cut from Vu to Wv is a straight chain of a legal c -edges in the second orientation. What remains for a uniform construction is precisely to fill across this change of grids; the exact 88-tiling does so for $(a, b, c) = (3, 5, 7)$.

APPENDIX E. RESULT LEDGER AND COMPUTATIONAL QUALIFICATIONS

Table 6 summarizes the results most relevant to a final solution. It does not claim that the displayed values form a complete frontier.

status	values or families	basis
members	the five classical families; published sporadic constructions; newly $88m^2$ and $189m^2$	classical theorems, cited constructions, exact certificates
nonmembers	7, 11, 14, 15; every prime $p > 3$ with $p \equiv 3 \pmod{4}$; and 21, 30, 33, 46	published proofs, proved arithmetic, or scoped exact searches
further audited exclusions	55, 56, 66, 70, 438	exact arithmetic audits; 56 and 70 include single-implementation exhaustive searches
finite-window ray problems	every Group 2 ray	Corollary 31
additive ray problems	Group 1 shapes 4–5; no cofinite tail is proved in general	Proposition 38
Group 2 rays with unresolved finite windows	for example the rays with base counts 33, 154, and 184	only finitely many multipliers remain undecided on each; the χ -based ideal-membership arithmetic is already exhausted

TABLE 6. Conservative status summary.

The table below intentionally omits the forced-layer-dependent exclusions. It also distinguishes “separately derived” from merely running two programs in the same repository: the 88 and 189 certificates have a checker independent of the search history, but a third-party reimplementations remains desirable.

The phrase “full branch audit” means that the classical, right-triangle, Group 1, Group 2, $\gamma = 2\alpha$, and 60° equilateral branches were all enumerated under the cited classification, rather than that one displayed congruence was treated as a global exclusion. The executable audits are in `computation/verify_theory_report3_claims.py` (including the explicit $N = 33$ reduction and the audits of 21, 30, 55, 66, 438) and `computation/sieve_120rows.py`; their commands and hashes are in `paper/verification_manifest.md`.

N	present basis for nonmembership	evidence
7, 11	Beeson’s direct proofs [1]	published
14, 15	Beeson’s finite equilateral analysis [1]; 14 also has a project exact exhaustion	published + search
$p \equiv 3 \pmod{4}$	three prime-case manuscripts [10, 9, 5]	independently derived
21	full branch audit; exact exhaustion of the surviving Group 1 shape-5 candidate, with an exported refutation certificate accepted by the independent checker	computer-assisted, independently certified
30	full branch audit; the Group 2 candidate fails boundary divisibility, with an exact exhaustion as a cross-check	arithmetic + search
33	unique candidate; exported refutation certificate accepted by the independent checker, corroborated by the primary exact exhaustion and by a replay disabling the general segment-semigroup table and boundary-invariant pruning	computer-assisted, independently certified
46	complete arithmetic branch audit; also [9]	independently derived
55	full branch audit; the Group 2 row-IV candidate has $b \nmid k$	proved + machine-audited arithmetic
56	full candidate audit; row-E and two shape-4 exact exhaustions	computer-assisted, decisive searches not independently ported
66	full branch audit; the Group 2 row-E candidate has $ab \nmid L$	proved + machine-audited arithmetic
70	full candidate audit; shape-5 exact exhaustion	computer-assisted, decisive search not independently ported
438	full arithmetic branch audit using the audited Group 1 shape-5 formula	machine-audited arithmetic

E.1. Dependency ledger. Tables 7 and 8 list every external result on which a conclusion of this paper depends, the version in which we read it, and its logical role. “Load-bearing” means that a stated conclusion of this paper becomes unproved if the imported statement fails; “corroboration” means that the conclusion has an independent basis here and the import is only a cross-check. Sources marked *unaudited in this project* were used as black boxes: we verified the statement we quote, not its proof.

Two structural remarks about this ledger. First, no conclusion of this paper depends on more than one unaudited preprint at a time: the Group-2 arithmetic depends on [7] alone, and the branch completeness on Laczkovich’s two papers as tabulated in [1] and on [4], but not on [6], which appears only as corroboration. Second, the two non-archival prime-case manuscripts and the private Beeson manuscript are corroboration only; every row that cites them has an independent basis in this paper or in an archival source.

APPENDIX F. PROVISIONAL BELOW-200 RESEARCH PROGRAMME

Not a theorem. Everything in this appendix depends on an unfinished forced-corner-layer induction and on a complete regeneration of the cross-branch candidate list. It is reported to direct future work and must not be cited as an established reduction.

The proposed forced-layer lemma would make a distinguished corner patch repeat along the equal sides of certain isosceles targets. The supporting line, chord partition, purity, and arithmetic residual steps have been checked. The missing step must exclude the mirror side-assignment of the tile placed in each interstitial gap and justify the endpoint-sector order. An undershooting mirror placement leaves angle-legal completions; an overshooting placement crosses the next chord line. No completed argument currently rules out every intermediate configuration.

A subsequent attempted proof correctly partitions the existing chord into coherently oriented tile edges and identifies the ideal triangular regions between two consecutive scaled patches. It then infers that the actual tiling occupies those ideal regions. That inference is precisely the missing step: the reflected congruent placement exchanges the two adjacent side lengths, undershoots one ideal wall at a T-junction, and crosses the artificial next chord on the other side. The pure-edge word theorem in Proposition 43 forces the residual angle at a coherent junction, but not this side assignment. Thus the later attempt does not complete the forced-layer induction or establish any of its consequences.

If the forced-layer induction and full candidate regeneration are completed as intended, the provisional ledger predicts that the unresolved values below 200 reduce to

105, 120, 132, 143, 154, 161, 175, 184.

At present, the status of at least

{60, 63, 76, 92, 99, 105, 120, 124, 132, 135, 140, 143, 154, 156,
161, 171, 172, 175, 184, 188}

is materially entangled with the missing induction or the audit built on it. In particular, neither “only eight values remain below 200” nor “every $N \leq 104$ is decided” is asserted here.

F.1. Provisional candidate table. Assuming the forced-layer induction and complete below-200 candidate regeneration, the provisional eight-value table is:

N	proposed surviving instance(s)
105	row IV: (8, 7, 13) at $k = 7$; (16, 5, 19) at $k = 5$
120	row IV: (7, 8, 13) at $k = 8$; a conditional 60° skeleton also remains
132	row I: (5, 3, 7) at multiplier 2
143	row V: (3, 5, 7) at multiplier 1
154	row I: (8, 7, 13) at multiplier 1
161	Group 1 shape 5: (12, 7, 16), $(p, q) = (3, 4)$, multiplier 1
175	$\gamma = 2\alpha$: (9, 7, 12), $(u, v) = (3, 4)$, $r = 5$
184	row I: (7, 8, 13) at multiplier 1

Positive transfer tests can be cheaper than direct searches: a row-IV (5, 3, 7) tiling at $N = 96$ would imply $132 \in \mathcal{S}$; the remaining Group 1 shape-4 instance at $N = 112$ would imply $161 \in \mathcal{S}$; and row-IV finds at 105 or 120 would also settle the linked values 154 or 184, respectively. These are one-way construction opportunities, not nonexistence proofs.

F.2. Priorities for deconditioning the ledger. The most direct next steps are:

- (1) close the forced-layer mirror gap with a side-assignment argument valid in the presence of T-junctions, together with the endpoint-sector order;
- (2) independently exhaust the $\gamma = 2\alpha$ instances at $N = 63$ and $N = 99$, and the 60° candidates at $N = 105$;
- (3) port the exact searcher to the $\gamma = 2\alpha$ geometry and attempt the sole proposed $N = 175$ instance; and
- (4) produce a script-generated all-branch candidate ledger through 200, with every imported theorem and conditional filter separately switchable.

DISCLAIMER: USE OF AI TOOLS

This project used multiple AI systems extensively. OpenAI GPT-5.6 Pro agents and Anthropic Claude Fable agents proposed lemmas, derived formulae, wrote and audited exact-search code, compared source versions, and helped draft this manuscript. Separate agents were assigned mathematical derivation, computation, source checking, and adversarial review. Agreement among models is an error filter, not independent scholarly verification. Every unconditional theorem is either proved in the text, imported with a precise citation, or explicitly labelled computer-assisted with its verification surface stated. Responsibility for any circulated version remains with the author.

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present conclusion	imported result	exact version	logical role
Completeness of the branch list, non-isosceles targets (Prop. 14(iv),(v),(xi)–(xiv))	Laczkovich, <i>Tilings of triangles</i> , Thm. 4.1, in the tabulation [1, Table 4]	Discrete Math. 140 (1995); tabulation arXiv:1811.09723v5	load-bearing; also fixes the target shape for the $\gamma = 2\alpha$ branch (case (ix))
same, restated	Beeson–Laczkovich–Zhang, <i>Solution of Erdős Problem 633</i> , Thm. 9	arXiv:2604.03609v2	corroboration; Thm. 9 verified verbatim against a local copy of the source, whose proof this project has not audited
Completeness, isosceles non-equilateral targets (Prop. 14(iii),(vi)–(viii),(x))	Beeson, Thm. 3(i)–(iii)	arXiv:2607.19572v1	load-bearing; its parts (i)–(iii) are attributed there to [3, Thm. 3.1] and [11]
Completeness, equilateral targets (Prop. 14(xv),(xvi))	Laczkovich, Thm. 3.3	Discrete Comput. Geom. 48 (2012)	load-bearing; read only through its use in the proof of [4, Thm. 11]; the 2012 wording was not re-verified here
Reptile counts (case (i))	Snover–Waiveris–Williams	Discrete Math. 91 (1991), as restated in [6, Thm. 4]	load-bearing
Commensurable-angle counts (case (ii))	Beeson, Thm. 3 and Table 3	arXiv:1811.09723v5	load-bearing; the last row of that table is flagged in the source as “necessary conditions for tilings that probably do not exist”
Right-tile classification (case (iii))	Beeson, Thm. 7.10	arXiv:1206.1974v7	load-bearing
Integer normalization of every Group-2 tile: Table 1, Cor. 22, Thm. 19	Beeson–Zhang, <i>Rationality of certain triangle tilings</i> , Thm. 1.2	arXiv:2604.01314v1, quoted verbatim as [15, Thm. 1] and as [6, Thm. 6]	load-bearing; statement verified through two independent quotations, source unaudited in this project
Integer normalization of every Group-1 tile: Table 4, Prop. 38	Beeson, Thm. 3 (“Rational tile”) and Lemma 2	arXiv:1206.2229v3	load-bearing; independent of arXiv:2604.01314v1
Normalization and parametrization of the $\gamma = 2\alpha$ tile: Prop. 40	Beeson, Thm. 10.5, Lemma 11.2, Lemma 11.4	arXiv:1206.1974v7	load-bearing for the parametrization and rationality; the area equation $X^2 = Nab$ is elementary

TABLE 7. Dependency ledger, part I: classification and normalization imports.

present conclusion	imported result	exact version	logical role
Shape-1 count $q^2(2q^2 - p^2)n^2$ used in Prop. 38 (Cor. 37)	Beeson, Thms. 4 and 6 (necessity) and Thm. 5 (sufficiency)	arXiv:1206.2229v3	load-bearing. We do <i>not</i> cite Thm. 7 of that paper, whose printed condition $K \nmid M^2$ is a typographical negation
Shape 1–3 characterizations and completeness of [2, Tables 1–3]	Beeson, Cor. 1, Table 2, Cor. 2	arXiv:1206.2229v3	corroboration for Table 2; the shape-3 ray of Table 4 additionally rests on the clause $g \mid M$ of [2, Thm. 14], which is not used in any exclusion here
Agreement of the surviving $N = 33$ instance with the published isosceles enumeration	Beeson, Thm. 12.10, Lemma 12.9, Table 5, Example 2	arXiv:1206.1974v7; verified locally in the michaelbeeson. com edition, numbering confirmed identical	corroboration; not used in the proof of Thm. 16. That source explicitly leaves $N = 33$ open and abstains from computer-search nonexistence proofs
Trapezoid and parallelogram constructions in Lemma 26	Zhang, Lemma 1 and Lemma 2 (Lemmas 24, 25)	arXiv:2512.22696v4	load-bearing; read with y meaning the leg, as that paper’s proofs do
Equilateral tail and its transfer to rows I, III, IV, II in Cor. 31	Zhang, Thm. 4, Props. 8–10, Prop. 11	arXiv:2512.22696v4	load-bearing as implications; their thresholds are replaced by the sharper window $m \geq 3R(a, b)$ proved here
Multipliers 2 and 4 on the Group-1 shape-5 ray $21n^2$ (Cor. 13)	Beeson, Table 5, rows $N = 84$ and $N = 336$ marked “yes”	arXiv:1206.2229v3	load-bearing: Cor. 13 needs multiplier 2. The negative entry at $n = 1$ is <i>not</i> imported; it is re-proved here
Orientation-changing block and the 720 inventory (Lemma 60, Rem. 44)	Beeson, Figs. 22 and 24 (Herdt’s constructions)	arXiv:1206.1974v7	corroboration; Rem. 44 audits a transcribed figure inventory and is not a coordinate certificate
Prime classification $p = 2, 3$ or $p \equiv 1 \pmod{4}$	Harries; George, GPT-5.6 Pro and Inkling; Beeson (all-prime); Beeson (isosceles prime)	as in the bibliography	corroboration; not used in any proof in this paper
Nonmembership of 46	George, GPT-5.6 Pro and Inkling	manuscript dated July 16, 2026	corroboration; not used in any proof in this paper. The basis here is the branch audit of <code>computation/ verify_theory_report3_ claims.py</code>
The clause $g \mid M$ of [2, Thm. 19]	Beeson, Thm. 19	arXiv:1206.2229v3	not used; see Remark 39. Its equation half is used only as a cross-check

TABLE 8. Dependency ledger, part II: counts, constructions, and corroborating sources.